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Preparation

15. How to Use the Front Panel

Setup of front panel lock

Outline

To prevent operational error e.g. unintentional parameter modification, the front panel may be locked.

Once locked, operations on the panel are limited as follows:

Mode	Locked panel conditions
Monitor Mode	No limitation: all monitored data can be checked.
Parameter Set up Mode	No parameter can be changed but setting can be checked.
EEPROM Writing Mode	Cannot be run. (No display)
Auxiliary Function Mode	Cannot be run except for "Release of front panel lock". (No display)

How to operate

• Related parameters

Parameter No.		Title	Function
Class	No.		
5	35	Setup of front panel lock	Locks the operation attempted from the front panel.

Lock and unlock can be made in one of two ways.

Procedure	Front panel	Setup support software PANATERM
Lock	(1) Set Pr5.35 "Front panel lock" to 1, and writ the setting to EEPROM. (2) Turn on power to the driver. (3) The front panel is locked.	
Unlock	(1) Execute the auxiliary function mode, front panel lock release function. (2) Turn on power to the driver. (3) The front panel is unlocked.	(1) Set Pr5.35 "Front panel lock" to 0, and writ the setting to EEPROM. (2) Turn on power to the driver. (3) The front panel is unlocked.

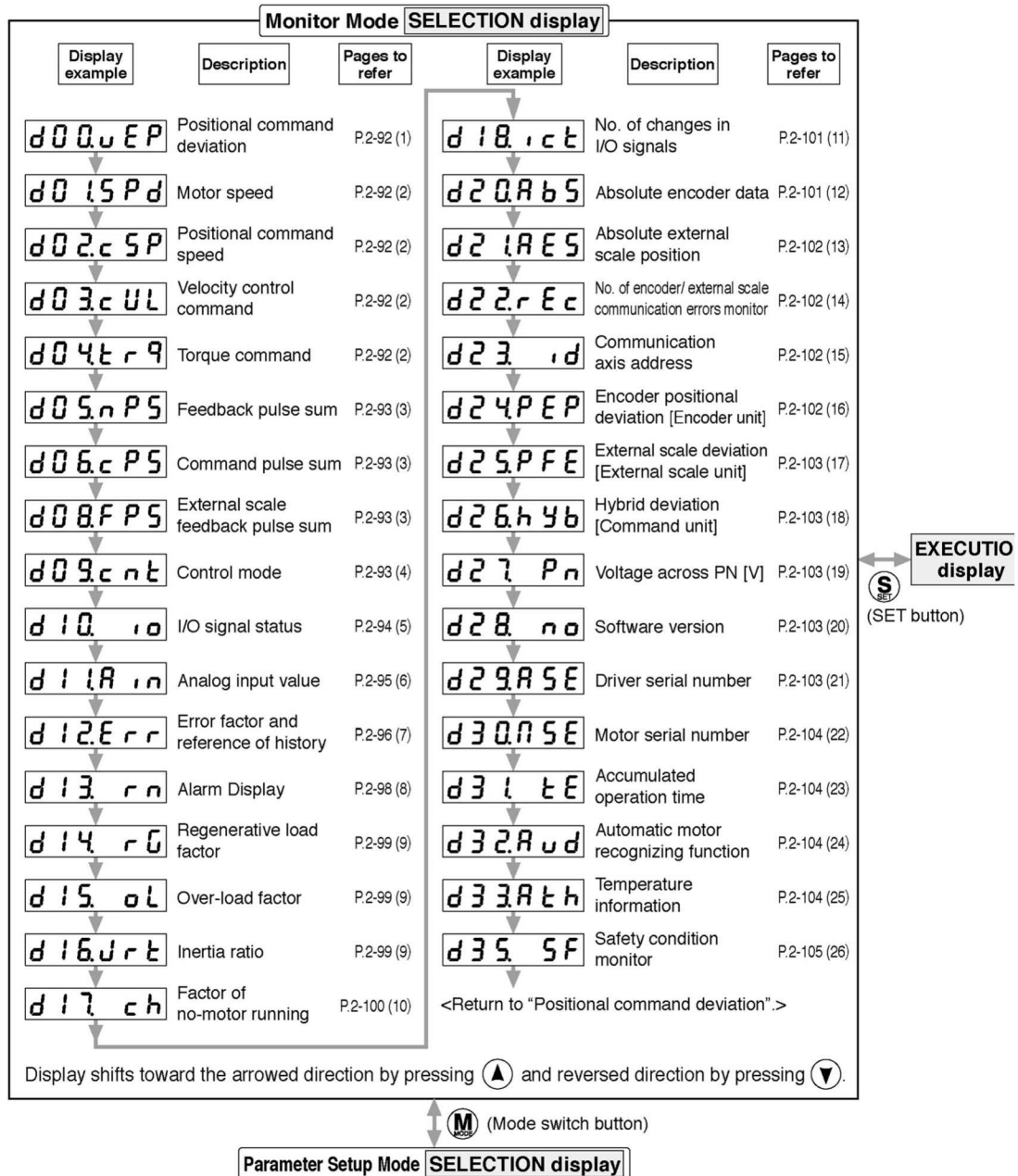
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Monitor Mode (SELECTION display)

To change the monitor display setting, select the display option to be changed from “**SELECTION**” display”, and press **(S)** to change to “**EXECUTION**” display”. After completion of changing, press **(S)** to return to the selection display,



Note

When you turn on the Product for the first time, display shows **r 0**. (at motor stall)
To change this display, change the setup of Pr5.28 (Initial status of LED).

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15. How to Use the Front Panel

Monitor Mode (EXECUTION display)

(1) Display of positional command deviation [command unit]

Displays positional deviation of the command unit in High order or Low order.

↑ Positional command deviation
LLow order
HHigh order

- To switch between Low order (L) and High order (H), press

(2) Display of motor speed, positional command speed, velocity control command and torque command

- Motor speed (r/min)

↑ Displays the motor speed (r/min).

- Positional command speed (r/min)

↑ Displays positional command speed (r/min).

- Velocity control command (r/min)

↑ Displays velocity control command (r/min).

- Torque command (%)

↑ Displays torque command (%).

15. How to Use the Front Panel

Monitor Mode (EXECUTION display)

(3) Display of Feedback Pulse Sum, Command Pulse Sum and External Scale Feedback Pulse Sum

- **Feedback Pulse Sum [Encoder feedback pulse]**

L 39025

A timing diagram showing a signal labeled "Feedback Pulse Sum". The signal is a square wave that transitions from a low state to a high state at a specific point in time, indicated by an upward arrow. The signal remains high for a duration, then transitions back to a low state, indicated by a downward arrow.

.....Low order

HHigh order

- To switch between Low order (L) and High order (H), press .

L39025 ↔ H 103

- **Command Pulse Sum [Command Pulse]**

L 3 9 0 2 5

.....Low order

HHigh order

- To switch between Low order (L) and High order (H), press .

L39025 ↔ H 103

• External Scale Feedback Pulse Sum

L 3 9 0 2 5

External Scale Feedback Pulse Sum

.....Low order

HHigh order

- To switch between Low order (L) and High order (H), press .

L39025 ↔ H 103

(4) Display of Control Mode

PoscntPosition control mode

SPdcntVelocity control mode

tr 9 c n tTorque control mode

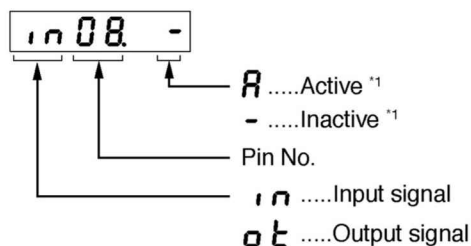
FclcntFull-closed control mode

15. How to Use the Front Panel

Monitor Mode (EXECUTION display)

(5) Display of I/O Signal Status

Displays the control input and output signal to be connected to connector X4.
Use this function to check if the wiring is correct or not.

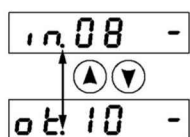


- Shift the flashing decimal point with .

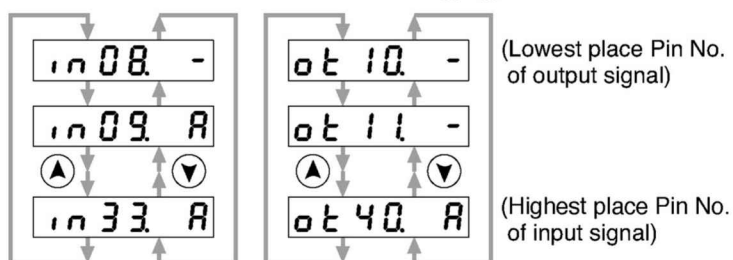
in 08 - (Right side of decimal point : Pin No. selection)

in 08 - (Left side of decimal point : Input/Output Pin No. selection)

- Select In or Out by pressing or button.



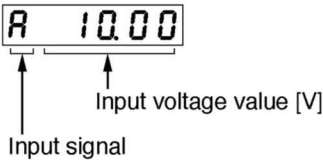
- Select the Pin No. to be monitored by pressing .



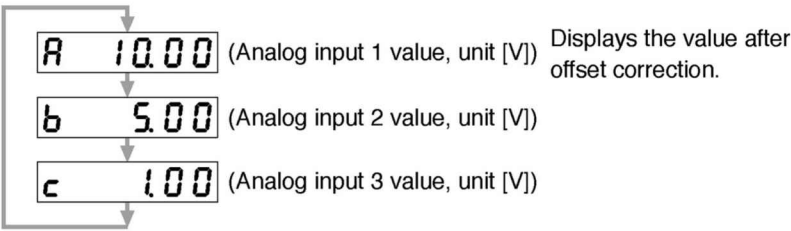
- *1 When input signal Active : Input signal photocoupler is ON.
 Inactive : Input signal photocoupler is OFF.
- When output signal Active : Output signal transistor is ON.
 Inactive : Output signal transistor is OFF.

15. How to Use the Front Panel
Monitor Mode (EXECUTION display)

(6) Display of Analog Input Value



•Select the signal No. to be monitored by pressings ▲ ▼.



Caution ⚠ Voltage exceeding $\pm 10V$ can not be displayed correctly.

15. How to Use the Front Panel

Monitor Mode (EXECUTION display)

(7) Display of Error Factor and Reference of History

Err.---

Error code No. (--- appears if no error occurs)

Err. Present error

E-0 History 0 (latest error)

E13 History 13 (oldest error)

• You can refer the last 14 error factors (including present one)
Press ▲ ▼ to select the factor to be referred.

<List of error code No.>

Error code		Protective function	Attribute		
Main	Sub		History	Can be cleared	Immediate stop
11	0	Control power supply under- voltage protection		○	
12	0	Over-voltage protection	○	○	
13	0	Main power supply under-voltage protection (between P to N)		○	
	1	Main power supply under-voltage protection (AC interception detection)		○	
14	0	Over-current protection	○		
	1	IPM error protection	○		
15	0	Over-heat protection	○		○
16	0	Over-load protection	○	○ ¹	
18	0	Over-regeneration load protection	○		○
	1	Over-regeneration Tr error protection	○		
21	0	Encoder communication disconnect error protection	○		
	1	Encoder communication error protection	○		
23	0	Encoder communication data error protection	○		
24	0	Position deviation excess protection	○	○	○
	1	Velocity deviation excess protection	○	○	○
25	0	Hybrid deviation excess error protection	○		○
26	0	Over-speed protection	○	○	○
	1	2nd over-speed protection	○	○	
27	0	Command pulse input frequency error protection	○	○	○
	2	Command pulse multiplier error protection	○	○	○
28	0	Limit of pulse replay error protection	○	○	○
29	0	Deviation counter overflow protection	○	○	
30	0	Safety detection		○	
33	0	IF overlaps allocation error 1 protection	○		
	1	IF overlaps allocation error 2 protection	○		
	2	IF input function number error 1 protection	○		
	3	IF input function number error 2 protection	○		
	4	IF output function number error 1 protection	○		
	5	IF output function number error 2 protection	○		
	6	CL fitting error protection	○		
	7	INH fitting error protection	○		

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Monitor Mode (EXECUTION display)

Error code		Protective function	Attribute		
Main	Sub		History	Can be cleared	Immediate stop
34	0	Software limit protection	○	○	
36	0 to 2	EEPROM parameter error protection			
37	0 to 2	EEPROM check code error protection			
38	0	Over-travel inhibit input protection		○	
39	0	Analog input1 excess protection	○	○	○
	1	Analog input2 excess protection	○	○	○
	2	Analog input3 excess protection	○	○	○
40	0	Absolute system down error protection	○	○	
41	0	Absolute counter over error protection	○		
42	0	Absolute over-speed error protection	○	○	
43	0	Initialization failure	○		
44	0	Absolute single turn counter error protection	○		
45	0	Absolute multi-turn counter error protection	○		
47	0	Absolute status error protection	○		
48	0	Encoder Z-phase error protection	○		
49	0	Encoder CS signal error protection	○		
50	0	External scale connection error protection	○		
	1	External scale communication error protection	○		
51	0	External scale status 0 error protection	○		
	1	External scale status 1 error protection	○		
	2	External scale status 2 error protection	○		
	3	External scale status 3 error protection	○		
	4	External scale status 4 error protection	○		
	5	External scale status 5 error protection	○		
55	0	A-phase connection error protection	○		
	1	B-phase connection error protection	○		
	2	Z-phase connection error protection	○		
87	0	Compulsory alarm input protection		○	
95	0 to 4	Motor automatic recognition error protection			
Other number		Other error	○		

Note

History...The error will be stored in the error history.

Can be cleared...To cancel the error, use the alarm clear input (A-CLR).

If the alarm clear input is not effective, turn off power, remove the cause of the error and then turn on power again.

Immediate stop...Instantaneous controlled stop upon occurrence of an error.

(Setting of "Pr.5.10 Sequence at alarm" is also required.)

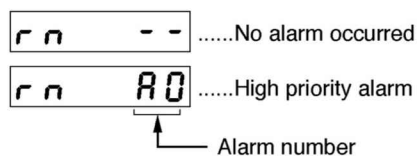
Caution

- 1) Certain alarms are not included in the history. For detailed information on alarms e.g. alarm numbers, refer to P.6-2.
- 2) When one of the errors which are listed in error history occurs, this error and history o shows the same error No.

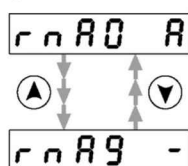
15. How to Use the Front Panel

Monitor Mode (EXECUTION display)

(8) Alarm Display



• To display the alarm occurrence condition, press $\blacktriangle$ or $\blacktriangledown$ button.



alarm No.	Alarm	Content	Latched time ^{*1}
A0	Overload protection	Load factor is 85% or more the protection level.	1 to 10s or ∞
A1	Over-regeneration alarm	Regenerative load factor is 85% or more the protection level.	10s or ∞
A2	Battery alarm	Battery voltage is 3.2 V or lower.	Fixed at ∞
A3	Fan alarm	Fan has stopped for 1 sec.	1 to 10s or ∞
A4	Encoder communication alarm	The number of successive encoder communication errors exceeds the specified value.	1 to 10s or ∞
A5	Encoder overheat alarm	The encoder detects overheat alarm.	1 to 10s or ∞
A6	Oscillation detection alarm	Oscillation or vibration is detected.	1 to 10s or ∞
A7	Lifetime detection alarm	Life expectancy of capacitor or fan is short.	Fixed at ∞
A8	External scale error alarm	The external scale detects the alarm.	1 to 10s or ∞
A9	External scale communication alarm	The number of successive external scale communication errors exceeds the specified value.	1 to 10s or ∞

^{*1} Alarms can be cleared by using the alarm clear. Because the all existing alarms are kept cleared while the alarm clear input (A-CLR) is kept ON, be sure to turn it OFF during normal operation. Either 1-10s or ∞ can be selected by using user parameter.

Exception: Battery alarm is fixed at ∞ because it is latched by the encoder.

Because the end of life alarm means that the life expectancy cannot be extended, the alarm is set at ∞ .

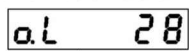
15. How to Use the Front Panel

Monitor Mode (EXECUTION display)

(9) Display of Regenerative Load Factor, Over-load Factor and Inertia Ratio**• Regenerative Load Factor**The image shows a digital display with two fields. The left field contains the characters 'rL' and the right field contains the number '30'. An arrow points from the text below to the '30' in the right field.

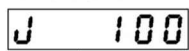
Display the ratio (%) against the alarm trigger level of regenerative protection.

This is valid when Pr0.16 (External regenerative resistor setup) is 0 or 1.

• Over-load FactorThe image shows a digital display with two fields. The left field contains the characters 'oL' and the right field contains the number '28'. An arrow points from the text below to the '28' in the right field.

Displays the ratio (%) against the rated load.

Refer to P.6-14, "Overload Protection Time Characteristics" of When in Trouble.

• Inertia RatioThe image shows a digital display with two fields. The left field contains the character 'J' and the right field contains the number '100'. An arrow points from the text below to the '100' in the right field.

Displays the inertia ratio (%).

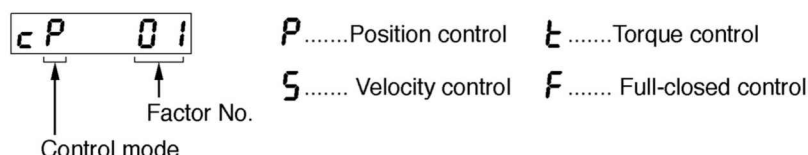
Value of Pr0.04 (Inertia Ratio) will be displayed as it is.

15. How to Use the Front Panel

Monitor Mode (EXECUTION display)

10) Display of the Factor of No-Motor Running

Displays the factor of no-motor running in number.



• Explanation of factor No.

Factor No.	Factor	Related Control Mode				Content
		P	S	T	F	
flashing	Occurrence of error/alarm	☐	☐	☐	☐	An error is occurring, and an alarm is triggered.
00	No particular factor	☐	☐	☐	☐	No factor is detected for No-motor run. The motor runs in normal case.
01	Main power shutoff	☐	☐	☐	☐	The main power of the driver is not turned on.
02	No entry of SRV-ON input	☐	☐	☐	☐	The Servo-ON input (SRV-ON) is not connected to COM-.
03	Over-travel inhibition input is valid	☐	☐	☐	☐	While Pr5.04 is 0 (Run-inhibition input is valid), • Positive direction over-travel inhibition input (POT) is open and speed command is Positive direction. • Negative direction over-travel inhibition input (NOT) is open and speed command is Negative direction.
04	Torque limit setup is small	☐	☐	☐	☐	Either one of the valid torque limit setup value of Pr0.13 (1st) or Pr5.22 (2nd) is set to 5% or lower than the rating.
05	Analog torque limit input is valid.	☐	☐	☐	☐	While Pr5.21 is 0 (analog torque limit input accepted), • Positive direction analog torque limit input (P-ATL) is negative voltage and speed command is Positive direction. • Negative direction analog torque limit input (N-ATL) is positive voltage and speed command is Negative direction.
06	INH input is valid.	☐	☐	☐	☐	Pr5.18 is 0 (Command pulse inhibition input is valid.), and INH is open.
07	Command pulse input frequency is low.	☐	☐	☐	☐	The position command per each control cycle is 1 pulse or smaller due to, • No correct entry of command pulse • No correct connection to the input selected with Pr0.05. • No matching to input status selected with Pr0.06 pr Pr0.07.
08	CL input is valid.	☐	☐	☐	☐	While Pr5.17 is 0 (Deviation counter clear at level), the deviation counter clear input (CL) is connected to COM-.
09	ZEROSPD input is valid.	☐	☐	☐	☐	While Pr3.15 is 1 (Speed zero clamp is valid.), the speed zero clamp input (ZEROSPD) is open.
10	External speed command is small.	☐	☐	☐	☐	While the analog speed command is selected, the analog speed command is smaller than 0.06[V].
11	Internal speed command is 0.	☐	☐	☐	☐	While the internal speed command is selected, the internal speed command is set to lower than 30 [r/min]
12	Torque command is small.	☐	☐	☐	☐	The analog torque command input (SPR or P-ATL) is smaller than 5 [%] of the rating.
13	Speed limit is small.	☐	☐	☐	☐	• While Pr3.17 is 0 (speed is limited by 4th speed of internal speed), Pr3.07, (4th speed of speed setup) is set to lower than 30 [r/min]. • While Pr3.17 is 1 (speed is limited by SPR input), the analog speed limit input (SPR) is smaller than 0.06 [V].
14	Other factor	☐	☐	☐	☐	The motor runs at 20 [r/min] or lower even though the factors from 1 to 13 are cleared, (the command is small, the load is heavy, the motor lock or hitting, driver/motor fault etc.)

Note * Motor might run even though the other number than 0 is displayed.

15. How to Use the Front Panel

Monitor Mode (EXECUTION display)

(11) Display of No. of changes in I/O signals

. 0 8 . 0

↑ ↑ ↑
 No. of changes in I/O signals (the signal is invalid)
 Pin No.
Input signal
 0Output signal

- Shift the flashing decimal point with ◀.

. 0 8 . 0

(Left side of decimal point : Pin No. selection)

. 0 / 8 . 0

(Right side of decimal point : Input/Output Pin No. selection)

- The switch of input/output, by pressing ▲ or ▼ button.

. 0 8 . 0

▲ ▼
 0 1 0 . 0

- Select the No. of pin, the number of changes on that pin should be displayed, by pressing ▲ or ▼ button.

. 0 8 . 0
 . 0 9 . 1
 ▲ ▼
 . 3 3 . 8

0 1 0 . 1
 0 1 1 . 0
 ▲ ▼
 0 4 0 . 3

(Lowest place Pin No. of output signal)

(Highest place Pin No. of input signal)

(12) Display of absolute encoder data

RL 38 10

↑ ↑ ↑
 Encoder data
 RLOne revolution data, Low order (L)
 RHOne revolution data, High order (H)
 bMulti-revolution data

- Select the data to be displayed by pressing ▲ or ▼ button.

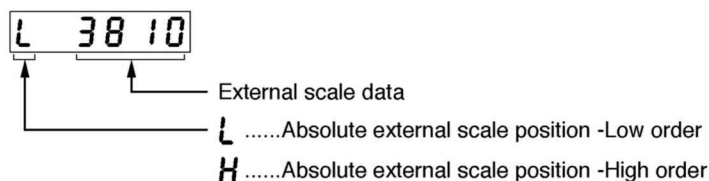
RL 38 10
 ▲ ▼
 RH 8
 ▲ ▼
 b 8

15. How to Use the Front Panel

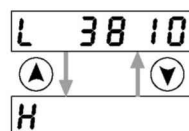
Monitor Mode (EXECUTION display)

(13) Display of absolute external scale position

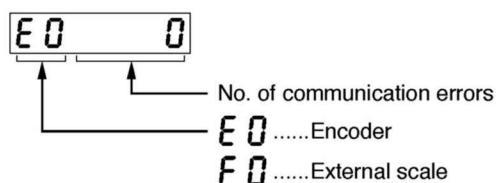
- Displays the absolute position of serial absolute scale.
- If a serial incremental scale, displays the scale position relative to the power on position which is defined as 0.



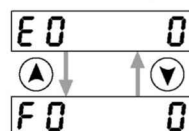
- Select encoder or external scale by pressing (▲) or (▼) button.



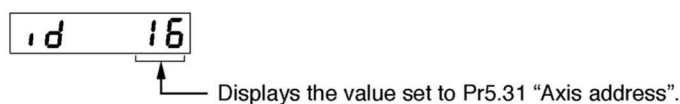
(14) Display of No. of encoder/ external scale communication errors monitor



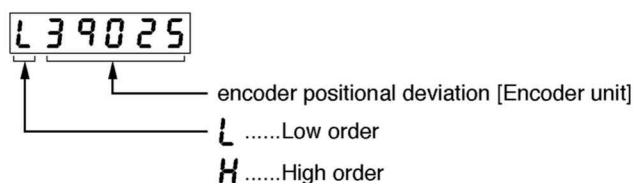
- Select encoder or external scale by pressing (▲) or (▼) button.



(15) Display of communication axis address



(16) Display of encoder positional deviation [Encoder unit]

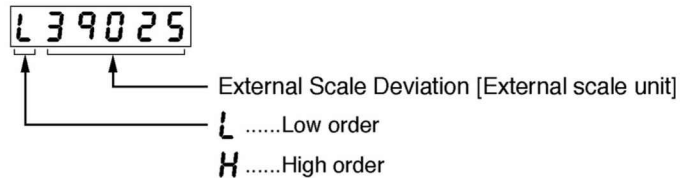


- To switch between Low order (L) and High order (H), press (◀).



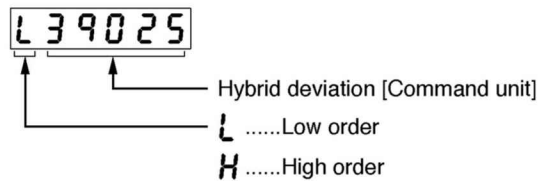
15. How to Use the Front Panel

Monitor Mode (EXECUTION display)

(17) Display of External Scale Deviation [External Scale Unit]

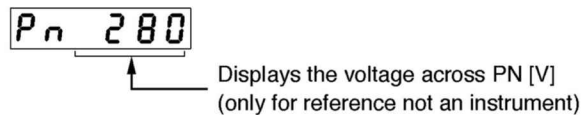
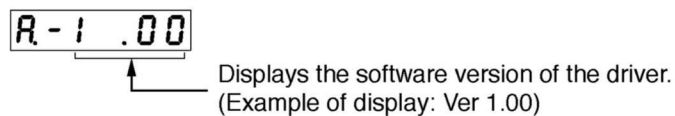
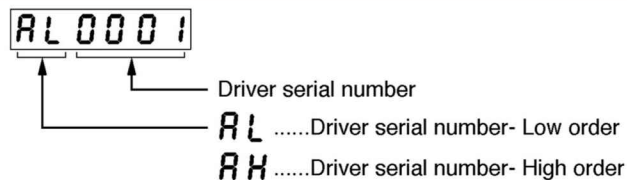
• To switch between Low order (L) and High order (H), press

L 39025 ↔ H 103

(18) Display of hybrid deviation [Command unit]

• To switch between Low order (L) and High order (H), press

L 39025 ↔ H 103

(19) Display of voltage across PN [V]**(20) Display of Software Version****(21) Display of driver serial number**

• To switch between Low order (L) and High order (H), press or .

(Example of display: Serial number 09010001)

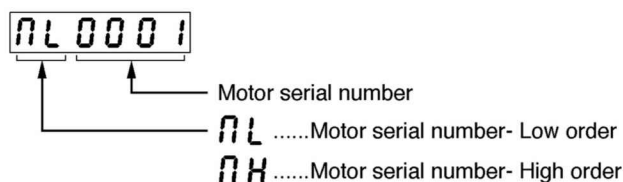
AL 0001

 AH 0904

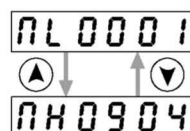
15. How to Use the Front Panel

Monitor Mode (EXECUTION display)

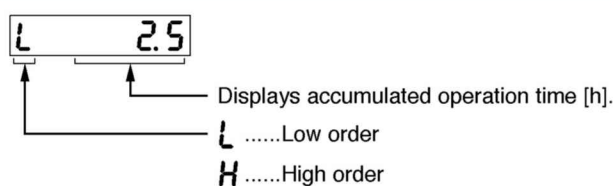
(22) Display of motor serial number



- To switch between Low order (L) and High order (H), press ▲ or ▼.
- (Example of display: Serial number 09040001)



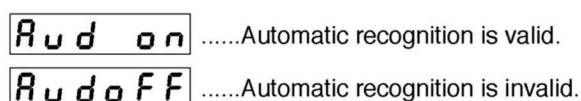
(23) Display of accumulated operation time



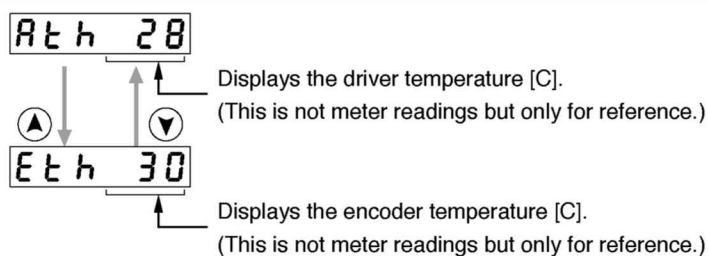
- To switch between Low order (L) and High order (H), press ◀.



(24) Automatic Motor Recognizing Function



(25) Display of temperature



(*) Can be other values than 500 depending on the ZERO/SPAN adjustment.

15. How to Use the Front Panel

Monitor Mode (EXECUTION display)

(26) Display of safety condition monitor

SAFETY

SAFETY : Safety condition
SrUoFF : Servo-off condition
SrUoN : Servo-on condition
ALArN : Alarm condition

Dot information

Flashing

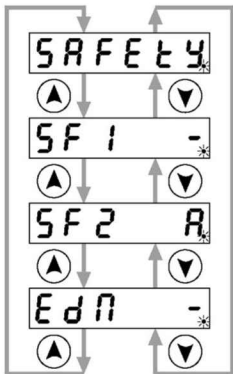
Normal change is possible

Servo ready condition

OFF: Dot unlit

ON: Dot lit

• Select desired monitor option by pressing ▲ or ▼ button.



- ...Input photocoupler OFF **A** ...Input photocoupler ON

- ...Input photocoupler OFF **A** ...Input photocoupler ON

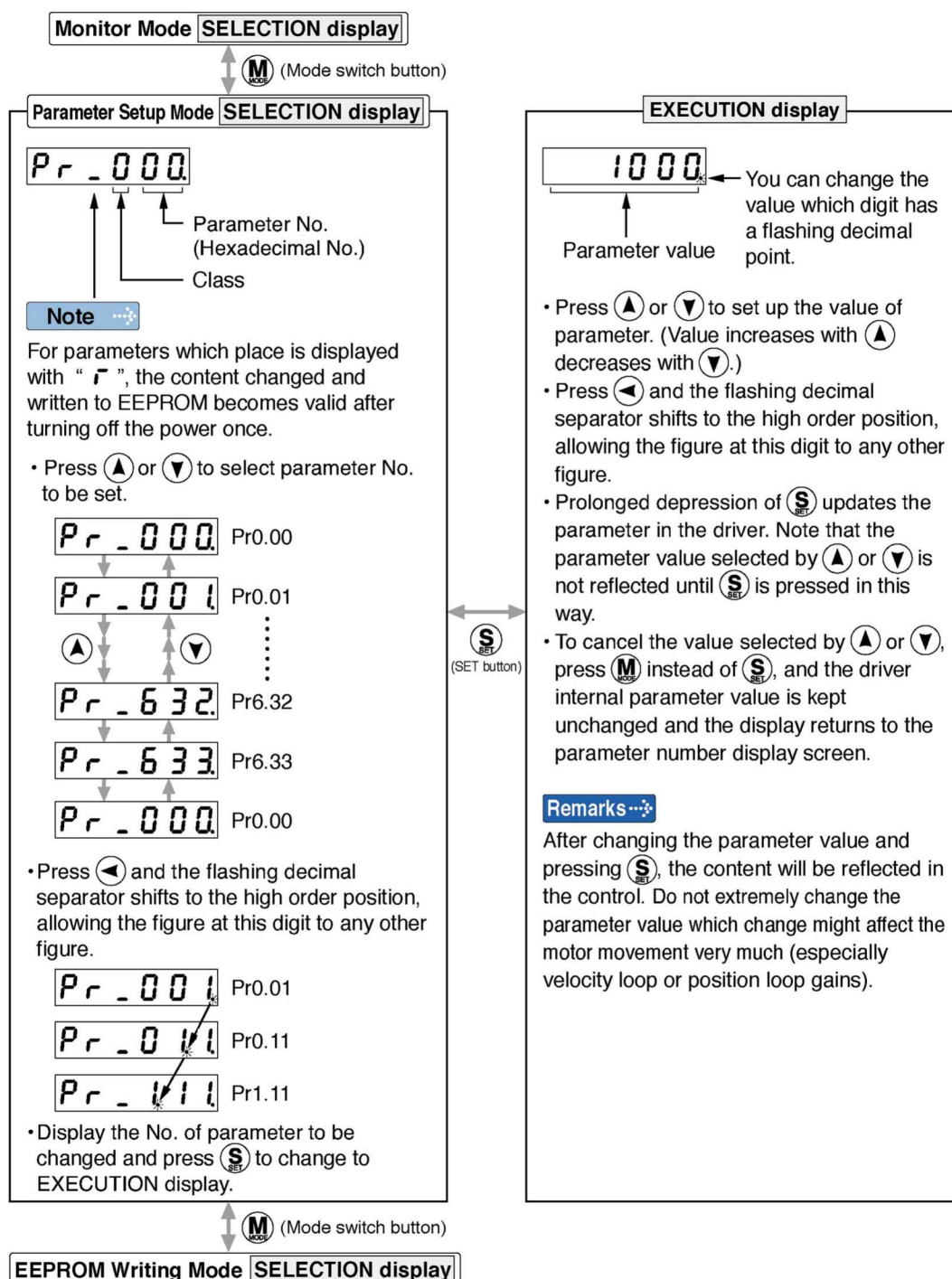
- ...Output photocoupler OFF **A** ...Output photocoupler ON

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Parameter Setup Mode

**Note**

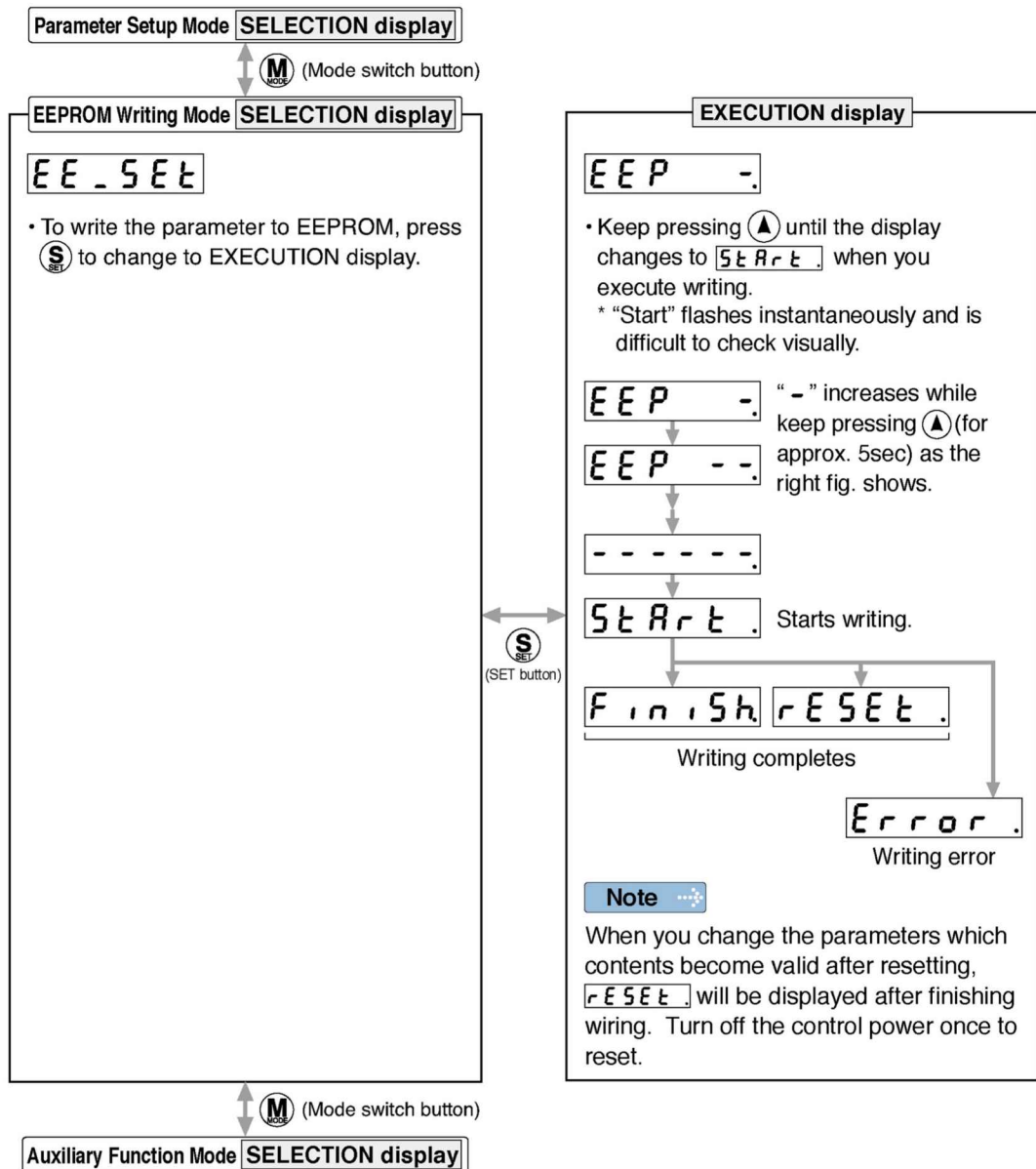
- After setting up parameters, return to SELECT mode, referring to structure of each mode (P.2-88).
- Each parameter has a limit in number of places for upper-shifting.

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15. How to Use the Front Panel

EEPROM Writing Mode

**Caution**

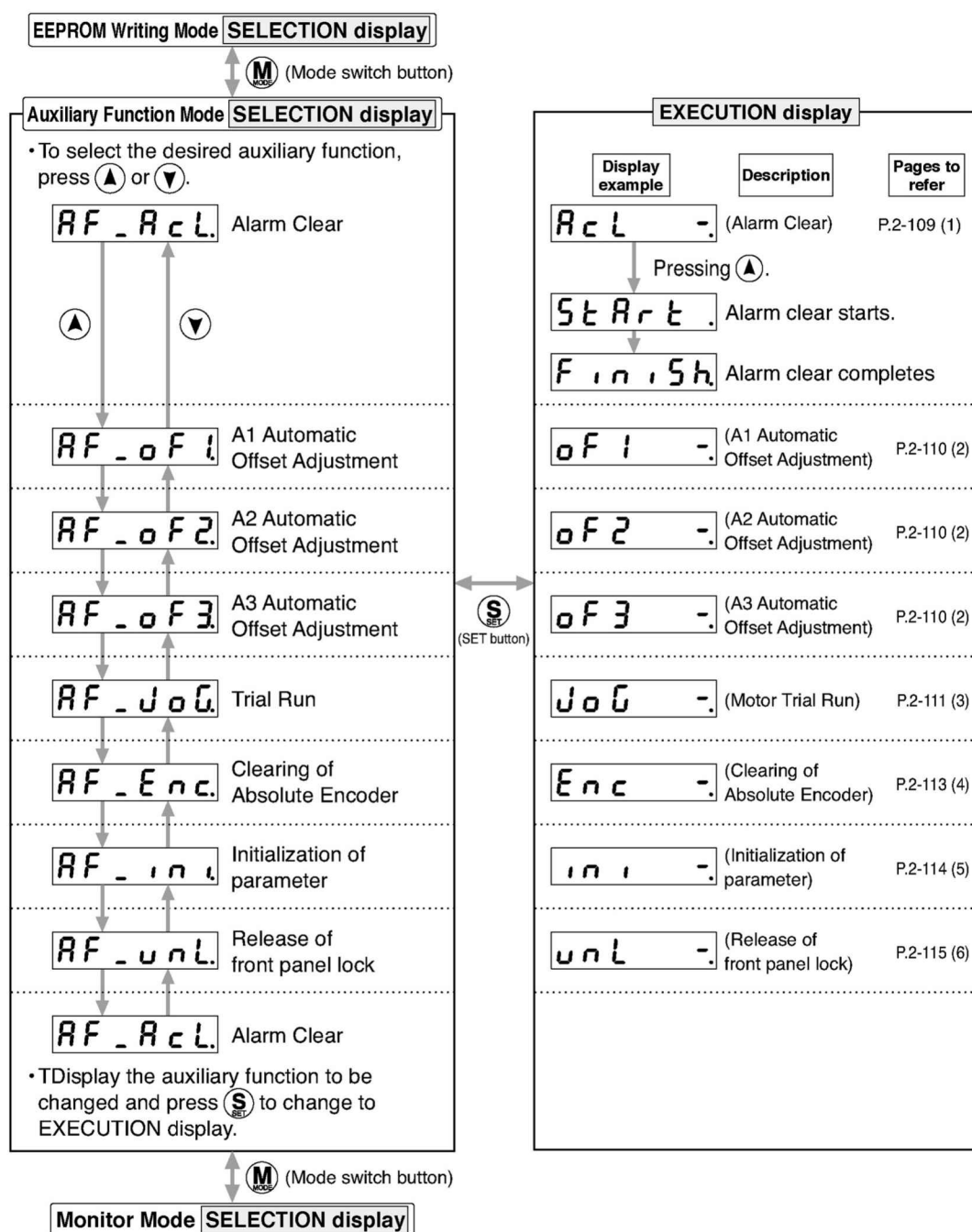
1. When writing error occurs, make writing again. If the writing error repeats many times, this might be a failure.
2. Don't turn off the power during EEPROM writing. Incorrect data might be written. If this happens, set up all of parameters again, and re-write after checking the data.
3. When the error defined by Err11.0 "Under voltage protection of control power supply" occurs, Error. is displayed indicating that no writing is made to EEPROM.

2

Preparation

15. How to Use the Front Panel

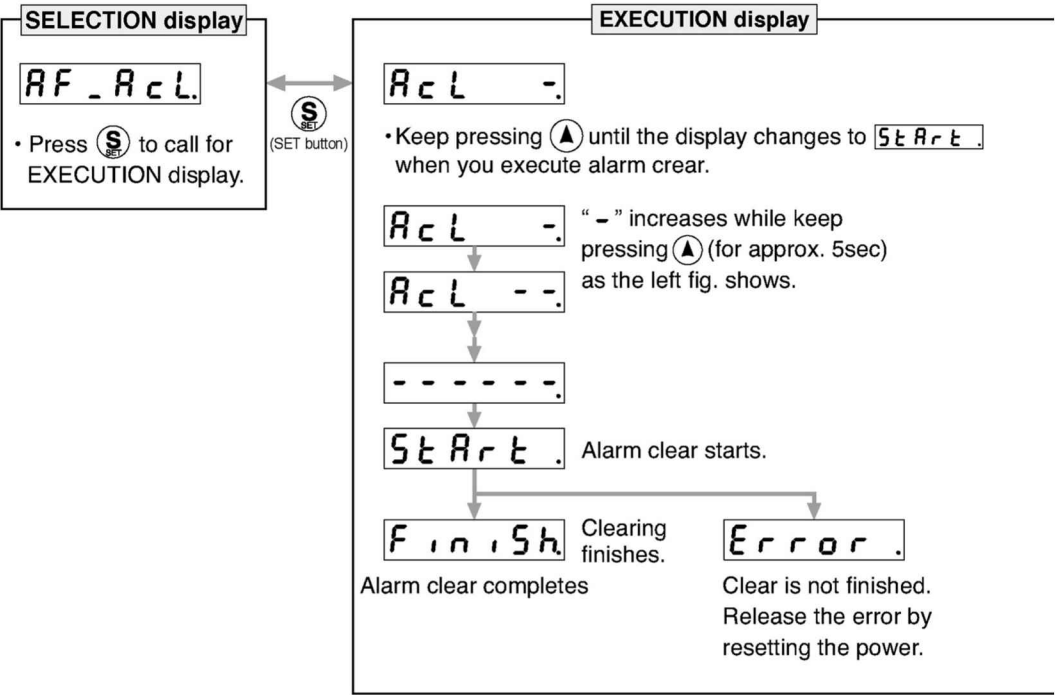
Auxiliary Function Mode (SELECTION display)



2 Preparation	15. How to Use the Front Panel
	Auxiliary Function Mode (EXECUTION display)

1) Alarm Clear Screen

This function releases the current alarm status.
Certain alarms will persist. If this is the case, refer to P.6-2 “When in Trouble - Protective Function”.



Note • After alarm cleaning, return to SELECTION display, referring to structure of each mode (P.2-88).

(*) Can be other values than 500 depending on the ZERO/SPAN adjustment.

15. How to Use the Front Panel

Auxiliary Function Mode (EXECUTION display)

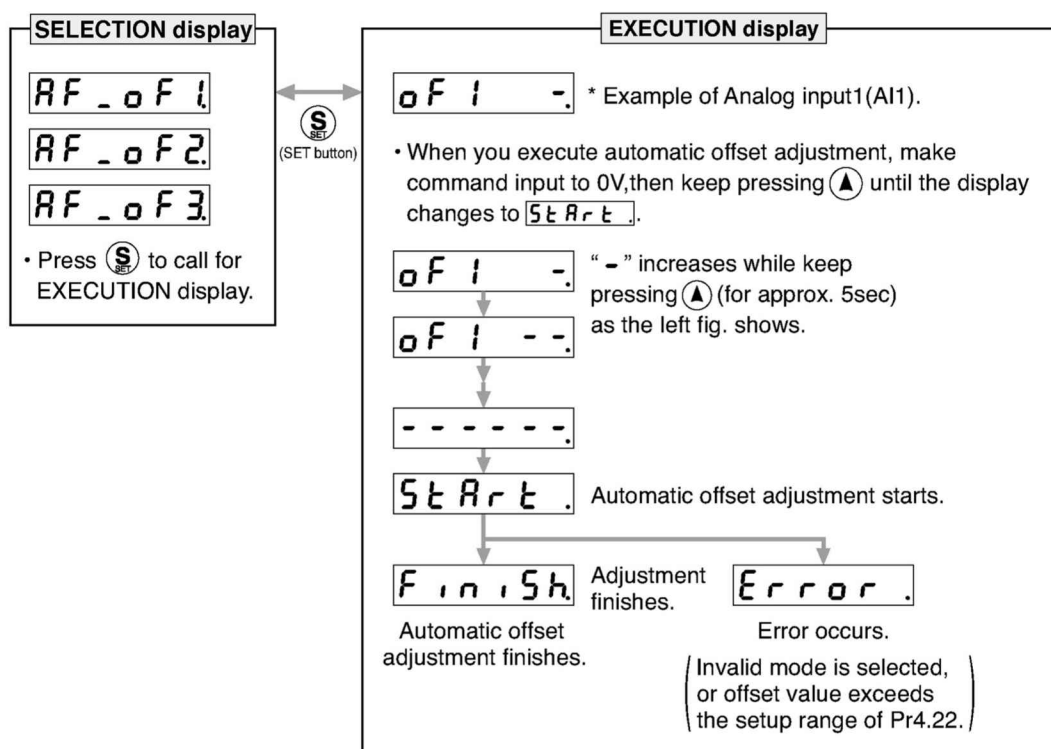
(2) Analog inputs 1 to 3 automatic offset adjustment

This function automatically adjusts offset setting of analog input.

Analog input 1 (AI1).....Pr4.22 (Analog input 1 (AI1) offset setup)

Analog input 2 (AI2).....Pr4.25 (Analog input 2 (AI2) offset setup)

Analog input 3 (AI3).....Pr4.28 (Analog input 1 (AI3) offset setup)



Remarks

- You cannot write the data only by executing automatic offset adjustment. Execute a writing to EEPROM when you need to reflect the result afterward.

Note

- After completion of the automatic offset adjustment, return to SELECTION display by referring to P.2-88 "Structure of Each Mode".

15. How to Use the Front Panel

Auxiliary Function Mode (EXECUTION display)

(3) Motor trial run

You can make a trial run (JOG run) without connecting the Connector, Connector X4 to the host controller such as PLC.

Remarks

- Separate the motor from the load, detach the Connector, Connector X4 before the trial run.
- Bring the user parameter setups (especially Pr0.04 and Pr1.01 to 1.04) to defaults, to avoid oscillation or other failure.

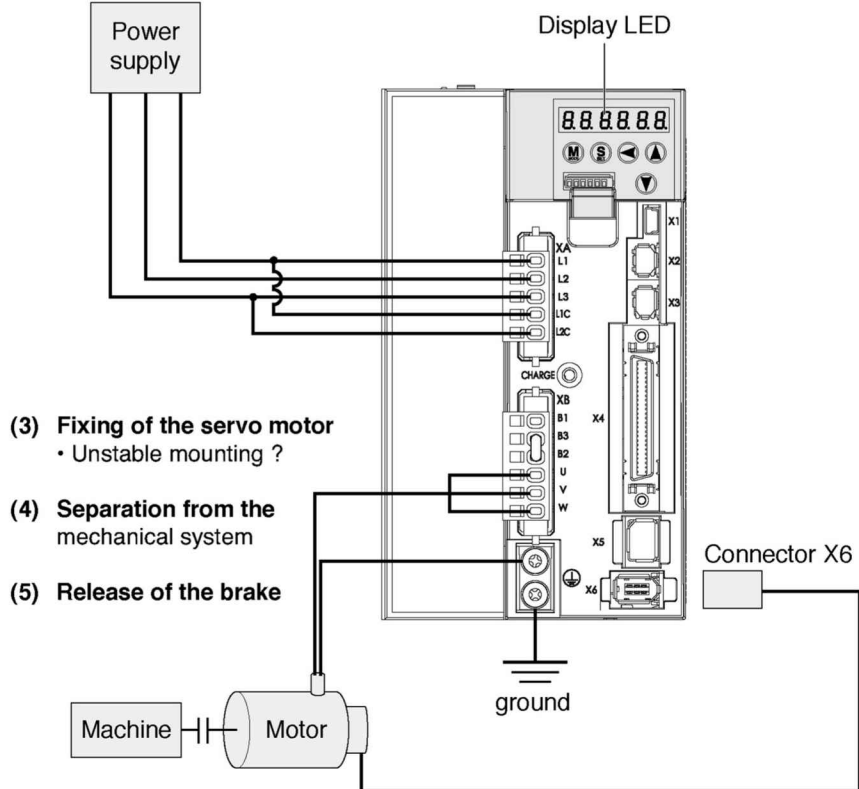
• Inspection Before Trial Run

(1) Inspection on wiring

- Miswiring ? (Especially power input and motor output)
- Short or grounded ?
- Loose connection ?

(2) Confirmation of power supply and voltage

- Rated voltage ?



(3) Fixing of the servo motor

- Unstable mounting ?

(4) Separation from the mechanical system

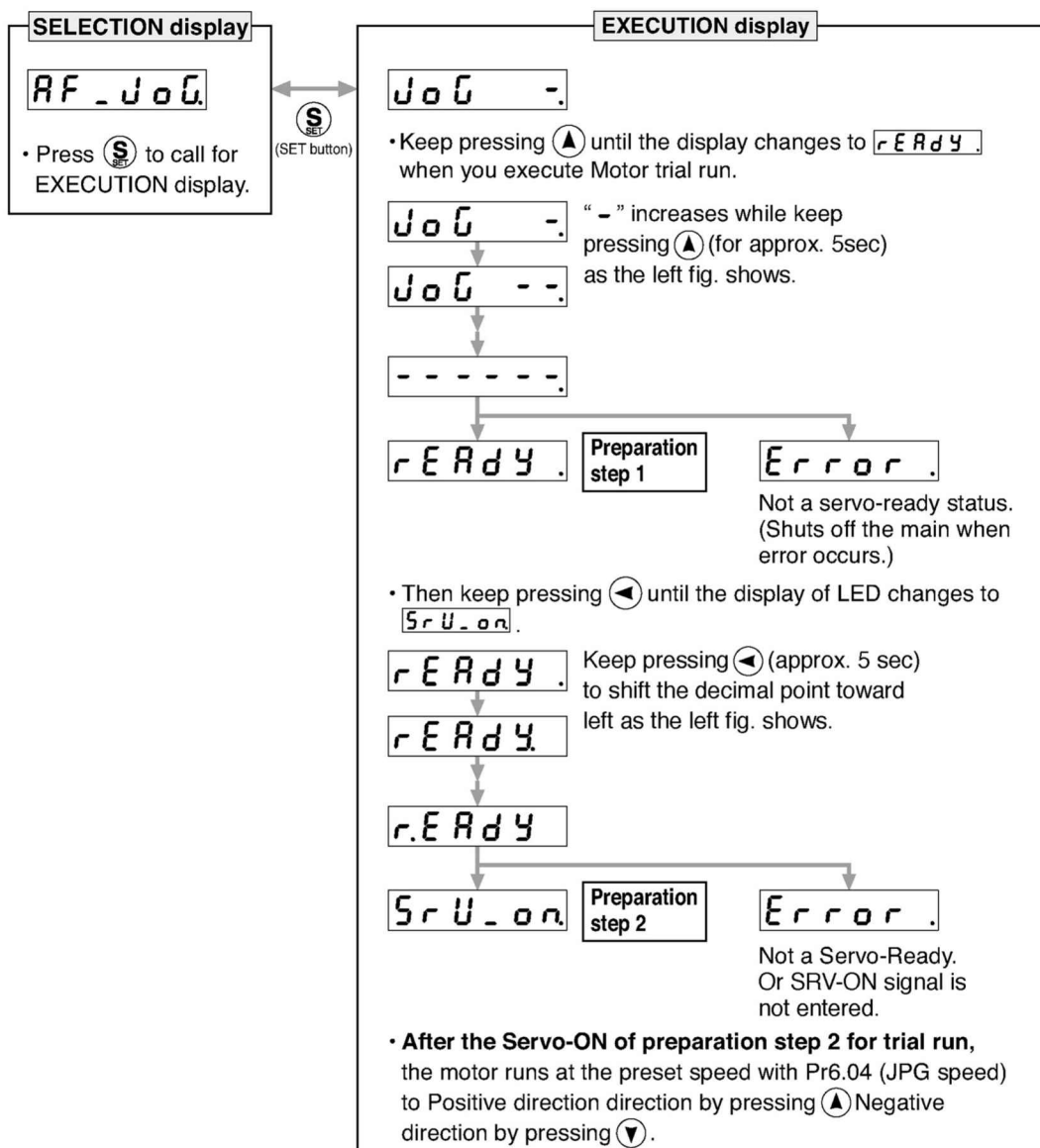
(5) Release of the brake

(6) Turn to Servo-OFF after finishing the trial run by pressing .

15. How to Use the Front Panel

Auxiliary Function Mode (EXECUTION display)

• Procedure for Trial Run



Caution

- Before starting the trial run, set the gain-related parameters to appropriate values to avoid problems such as oscillation. If the load is removed, be sure to set Pr0.04 “Inertia Ratio” to 0.
- During the trial run, use the velocity control mode. Various settings including parameters should assure safe and positive operation under appropriate velocity control.
- If SRV-ON becomes valid during trial run, the display changes to **Error**, which is normal run through external command.

Note

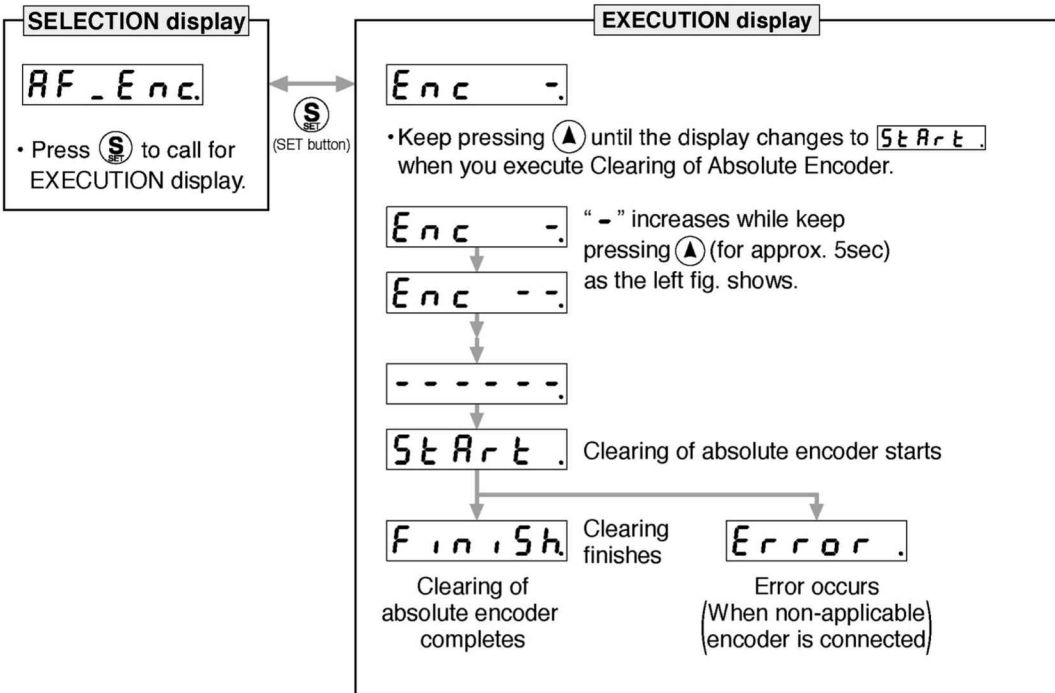
- After finished trial running, return to SELECTION display, referring to structure of each mode (P.2-88).

15. How to Use the Front Panel

Auxiliary Function Mode (EXECUTION display)

4) Clearing of Absolute Encoder

You can clear the multi-turn data of the absolute encoder.



Note

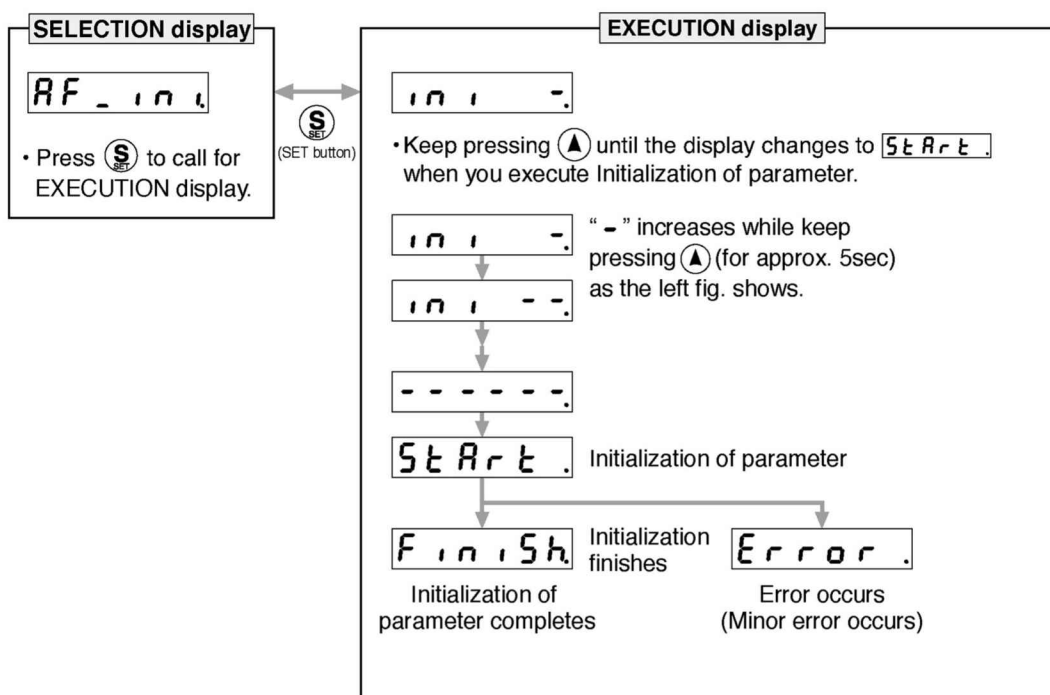
• After clearing of absolute encoder finishes, return to SELECTION display, referring to structure of each mode (P.2-88).

15. How to Use the Front Panel

Auxiliary Function Mode (EXECUTION display)

(5) Initialization of parameter

Initialize the parameter.



Caution • Parameter cannot be initialized when one of the following error occurs: Err11.0 "Under voltage protection of control power supply", EEPROM related errors (Err36.0, Err36.1, Err36.2, Err37.0, Err37.1 and Err37.2) - initialization will result in "Error" display.

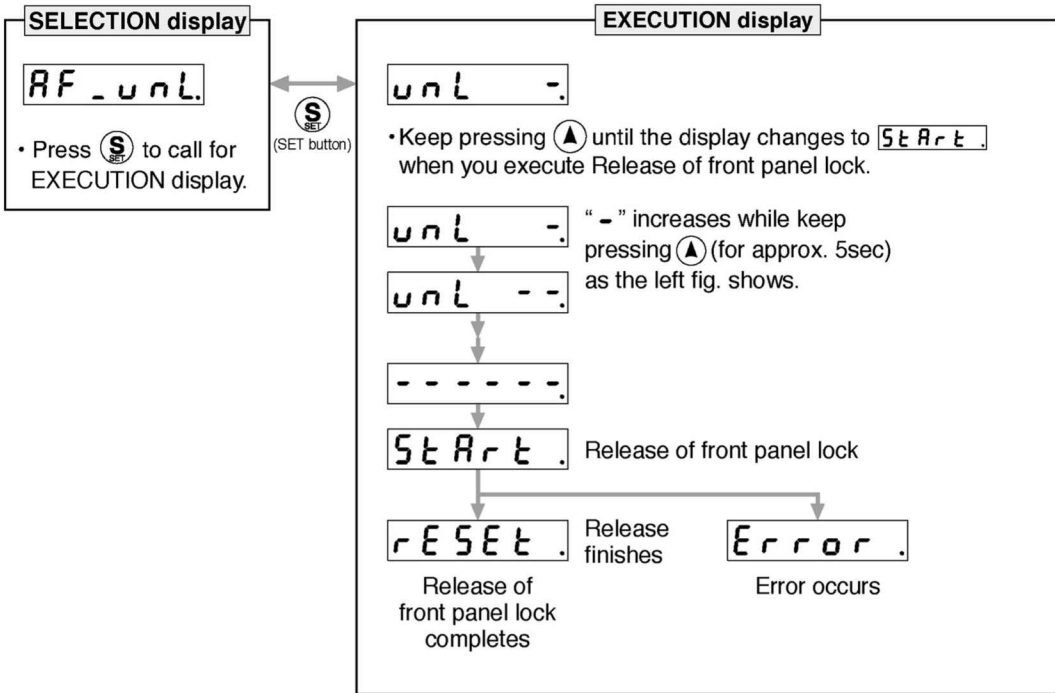
Note • After initialization of parameter finishes, return to SELECTION display, referring to structure of each mode (P.2-88).

15. How to Use the Front Panel

Auxiliary Function Mode (EXECUTION display)

(6) Release of front panel lock

Release the front panel lock setting.



Note

• After release of front panel lock finishes, return to SELECTION display, referring to structure of each mode (P.2-88).

ATLAS-204

- Class Pr0.00 Parameter No.

[Class 0] Basic setting[Class 1] Gain adjustment

Parameter No.	Title		M1 (PullDown)	M2 (Jaw Rotation)	M3 (Pivot)	M4 (Film Roll)	M5 (BackSeal)	M7 (Shaker)	M9 (Poker)
Class	No.								
[Class 2] Damping control	00	Adaptive filter mode setup	1	1	1	1	1	1	4
	01	1st notch	frequency	1000	5000	5000	5000	5000	5000
	02		width selection	2	2	2	2	2	2
	03		depth selection	0	0	0	0	0	0
	04	2nd notch	frequency	1000	1500	1500	1000	1000	5000
	05		width selection	2	2	2	2	2	2
	06		depth selection	0	0	0	0	0	0
	07	3rd notch	frequency	5000	562	562	5000	5000	2456
	08		width selection	2	2	2	2	2	2
	09		depth selection	0	0	0	0	0	0
	10	4th notch	frequency	5000	5000	5000	5000	5000	5000
	11		width selection	2	2	2	2	2	2
	12		depth selection	0	0	0	0	0	0
	13	Selection of damping filter switching	0	0	0	0	0	0	0
	14	1st damping	frequency	0	0	0	0	0	0
	15		filter setup	0	0	0	0	0	0
	16	2nd damping	frequency	0	0	0	0	0	0
	17		filter setup	0	0	0	0	0	0
	18	3rd damping	frequency	0	0	0	0	0	0
	19		filter setup	0	0	0	0	0	0
	20	4th damping	frequency	0	0	0	0	0	0
	21		filter setup	0	0	0	0	0	0
	22	Positional command	smoothing filter	2	2	2	2	2	2
	23		FIR filter	0	0	0	0	0	0
[Class 3] Velocity/Torque/Full-closed control	00	Speed setup, Internal/External switching	0	0	0	0	0	0	0
	01	Speed command	rotational direction selection	0	0	0	0	0	0
	02		input gain	500(*)	200	400	500(*)	500(*)	300
	03		reversal input	1	0	0	1	0	1
	04	Speed setup	1st	0	0	0	0	0	0
	05		2nd	0	0	0	0	0	0
	06		3rd	0	0	0	0	0	0
	07		4th	0	0	1200	0	0	0
	08		5th	0	0	0	0	0	0
	09		6th	0	0	0	0	0	0
	10		7th	0	0	0	0	0	0
	11		8th	0	0	0	0	0	0
	12	Time setup	acceleration	0	0	0	0	0	0
	13		deceleration	0	0	0	0	0	0
	14	Sigmoid acceleration/deceleration time setup	0	0	0	0	0	0	0
	15	Speed zero-clamp	function selection	1	0	0	1	1	1
	16		level	30	30	30	30	30	30
	17	Torque command	selection	0	0	0	0	0	0
	18		direction selection	0	0	0	0	0	0
	19		input gain	30	30	30	30	30	30
	20		input reversal	0	0	0	0	0	0
	21	Speed limit value	1	0	0	0	0	0	0
	22		2	0	0	0	0	0	0
	23	External scale	selection	2	2	2	2	2	2
	24		numerator of division	0	0	0	0	0	0
	25		denominator of division	10000	10000	10000	10000	10000	10000
	26		reversal of direction	0	0	0	0	0	0
	27		Z phase disconnection detection disable	0	0	0	0	0	0
	28	Hybrid deviation	excess setup	122	1600	1600	122	122	1600
	29		clear setup	0	0	0	0	0	0

Parameter No.	Title		M1 (PullDown)	M2 (Jaw Rotation)	M3 (Pivot)	M4 (Film Roll)	M5 (BackSeal)	M7 (Shaker)	M9 (Poker)
Class	No.								
[Class 4] I/F monitor setting	00	Input selection	SI1 (Pin No.8)	8553090	8553090	8553090	8553090	8553090	8553090
	01		SI2 (Pin No.9)	8487297	8487297	8487297	8487297	8487297	8487297
	02		SI3 (Pin No.26)	9539850	9539850	9539850	9539850	9539850	9539850
	03		SI4 (Pin No.27)	394758	394758	394758	394758	394758	394758
	04		SI5 (Pin No.28)	4108	4108	4108	4108	4108	4108
	05		SI6 (Pin No.29)	197379	197379	197379	197379	197379	197379
	06		SI7 (Pin No.30)	3847	3847	3847	3847	3847	3847
	07		SI8 (Pin No.31)	263172	263172	263172	263172	263172	263172
	08		SI9 (Pin No.32)	328965	328965	328965	328965	328965	328965
	09		SI10 (Pin No.33)	3720	3720	3720	3720	3720	3720
	10	Output selection	SO1 (Pin No.10, 11 Line driver output)	197379	197379	197379	197379	197379	197379
	11		SO2 (Pin No.34, 35 Line driver output)	131586	131586	131586	131586	131586	131586
	12		SO3 (Pin No.36, 37 Line driver output)	65793	65793	65793	65793	65793	65793
	13		SO4 (Pin No.38, 39 Line driver output)	328964	328964	328964	328964	328964	328964
	14		SO5 (Pin No.12 Open collector output)	460551	460551	460551	460551	460551	460551
	15		SO6 (Pin No.40 Open collector output)	394758	394758	394758	394758	394758	394758
	16	Analog monitor 1	type	0	0	0	0	0	0
	17		output gain	500	500	500	500	500	500
	18	Analog monitor 2	type	4	4	4	4	4	4
	19		output gain	33	33	33	33	33	33
	20	Type of digital monitor		0	0	0	0	0	0
	21	Analog monitor output setup		0	0	0	0	0	0
	22	Analog input 1 (AI1)	offset setup	0	0	0	0	0	0
	23		filter	0	0	0	0	0	0
	24		overvoltage setup	0	0	0	0	0	0
	25	Analog input 2 (AI2)	offset setup	0	0	0	0	0	0
	26		filter	0	0	0	0	0	0
	27		overvoltage setup	0	0	0	0	0	0
	28	Analog input 3 (AI3)	offset setup	0	0	0	0	0	0
	29		filter	0	0	0	0	0	0
	30		overvoltage setup	0	0	0	0	0	0
	31	Positioning complete (In-position)	range	1	10	10	1	1	10
	32		output setup	0	0	0	0	0	0
	33	INP hold time		0	0	0	0	0	0
	34	Zero-speed		50	50	50	50	50	50
	35	Speed coincidence range		50	50	50	50	50	50
	36	At-speed (Speed arrival)		1000	1000	1000	1000	1000	1000
	37	Mechanical brake action	at stalling setup	0	0	0	0	0	0
	38		at running setup	0	0	0	0	0	0
	39	Mechanical brake action at running setup		30	30	30	30	30	30
	40	Selection of alarm output	1	0	0	0	0	0	0
	41		2	0	0	0	0	0	0
	42	2nd Positioning complete (In-position) range		10	10	10	10	10	10

Parameter No.	Title		M1	M2	M3	M4	M5	M7	M9
Class	No.		(PullDown)	(Jaw Rotation)	(Pivot)	(Film Roll)	(BackSeal)	(Shaker)	(Poker)
[Class 5] Enhancing setting	00	2nd	numerator of electronic gear	0	0	0	0	0	0
	01	3rd	numerator of electronic gear	0	0	0	0	0	0
	02	4th	numerator of electronic gear	0	0	0	0	0	0
	03	Denominator of pulse output division		0	0	0	0	0	0
	04	Over-travel inhibit input setup		1	1	1	1	1	1
	05	Sequence at over-travel inhibit		0	0	0	0	0	0
	06	Sequence at Servo-Off		3	0	0	3	3	3
	07	Main power OFF	sequence	3	0	0	3	3	3
	08		LV trip selection	1	1	1	1	1	1
	09		detection time	70	70	70	70	70	70
	10	Sequence at alarm		3	0	0	3	3	3
	11	Torque setup for emergency stop		0	0	0	0	0	0
	12	Over-load level setup		0	0	0	0	0	0
	13	Over-speed level setup		0	0	0	0	0	0
	14	Motor working range setup		10	10	10	10	10	10
	15	I/F reading filter		0	0	0	0	0	0
	16	Alarm clear input setup		0	0	0	0	0	0
	17	Counter clear input mode		3	3	3	3	3	3
	18	Command pulse inhibit input	invalidation	1	1	1	1	1	1
	19		reading setup	4	4	4	4	4	4
	20	Position setup unit select		0	0	0	0	0	0
	21	Selection of torque limit		1	1	1	1	1	1
	22	2nd torque limit		300	300	300	300	300	300
	23	Torque limit switching setup	1	0	0	0	0	0	0
	24		2	0	0	0	0	0	0
	25	External input	positive direction torque limit	300	300	500	300	300	500
	26		negative direction torque limit	300	300	500	300	300	500
	27	Input gain of analog torque limit		30	30	30	30	30	30
	28	LED initial status		1	1	1	1	1	1
	29	RS232	baud rate setup	2	2	2	2	2	2
	30	RS485		2	2	2	2	2	2
	31	Axis address		1	1	1	1	1	1
	32	Command pulse input maximum setup		4000	4000	4000	4000	4000	4000
	33	Pulse regenerative output limit setup		0	0	0	0	0	0
	34	For manufacturer's use		4	4	4	4	4	4
	35	Front panel lock setup		0	0	0	0	0	0

Parameter No.		Title	M1 (PullDown)	M2 (Jaw Rotation)	M3 (Pivot)	M4 (Film Roll)	M5 (BackSeal)	M7 (Shaker)	M9 (Poker)
Class	No.								
[Class 6] Special setting	00	Analog torque feed forward conversion gain	0	0	0	0	0	0	0
	02	Velocity deviation excess setup	0	0	0	0	0	0	0
	04	JOG trial run command speed	300	300	300	300	300	300	50
	05	Position 3rd gain	0	0	0	0	0	0	0
	06		100	100	100	100	100	100	100
	07	Torque command additional value	0	0	0	0	0	0	0
	08	Positive direction	0	0	0	0	0	0	0
	09	Negative direction	0	0	0	0	0	0	0
	10	Function expansion setup	0	0	0	0	0	0	0
	11	Current response setup	100	100	100	100	100	100	100
	13	2nd Inertia ratio	250	250	250	250	250	250	250
	14	Emergency stop time at alarm	200	200	200	200	200	200	200
	15	2nd over-speed level setup	0	0	0	0	0	0	0
	17	Front panel parameter writing selection	0	0	0	0	0	0	0
	18	Power-up wait time	0	0	0	0	0	0	0
	19	Encoder Z phase setup	0	0	0	0	0	0	0
	20	Z-phase setup of external scale	0	0	0	0	0	0	0
	21	Serial absolute external scale Z phase setup	0	0	0	0	0	0	0
	22	A, B phase external scale pulse output method selection	0	0	0	0	0	0	0
	23	Disturbance torque compensating gain	0	0	0	0	0	0	0
	24	Disturbance observer filter	53	53	53	53	53	53	53
	27	Alarm latch time selection	1	5	5	1	1	1	1
	31	Real time auto tuning	1	1	1	1	1	1	1
	32		0	0	0	0	0	0	0
	34	Hybrid vibration suppression	0	0	0	0	0	0	0
	35		10	10	10	10	10	10	10
	37	Oscillation detection level	0	0	0	0	0	0	0
	38	Alarm mask setup	0	0	4	0	0	0	0
	39	For manufacturer's use	0	0	0	0	0	0	0

ATLAS-234

• A parameter is designated as follows:

Pr0.00
 Class ——— Parameter No.

• Definition of symbols under “Related mode”

P: position control, S: velocity control,
 T: torque control, F: full closed control

Parameter No.	Title		M1 (PullDown)	M2 (Jaw Rotation)	M3 (Pivot)	M4 (Film Roll)	M5 (BackSeal)	M7 (Shaker)	M9 (Poker)
Class	No.								
[Class 0] Basic setting	00	Rotational direction setup	1	1	1	1	1	1	1
	01	Control mode setup	1	1	5	1	1	1	
	02	Real-time auto-gain tuning	0	0	0	0	0	0	
	03	selection of machine stiffness	12	11	11	12	12	12	12
	04	Inertia ratio	100	0	0	100	100	100	0
	05	Command pulse	input selection	0	0	0	0	0	0
	06		rotational direction setup	0	0	0	0	0	0
	07		input mode setup	1	1	1	1	1	1
	08	Command pulse counts per one motor revolution	10000	10000	10000	10000	10000	10000	10000
	09	1st numerator of electronic gear	0	0	0	0	0	0	0
	10	Denominator of electronic gear	10000	10000	10000	10000	10000	10000	10000
	11	Output pulse counts per one motor revolution	1000	2500	1000	2500	2500	2500	250
	12	Reversal of pulse output logic	0	0	0	0	0	0	0
	13	1st torque limit	300	300	250	300	300	300	300
	14	Position deviation excess setup	488281	6400000	6400000	488281	488281	488281	6400000
	15	Absolute encoder setup	1	1	1	1	1	1	1
	16	External regenerative resistor setup	0	1	0	3	3	3	0
	17	Load factor of external regenerative resistor selection	0	0	0	0	0	0	0
[Class 1] Gain adjustment	00	1st	gain of position loop	500	500	500	500	500	500
	01		gain of velocity loop	1000	1000	2000	1000	1000	500
	02		time constant of velocity loop integration	500	10000	500	500	500	500
	03		filter of speed detection	4	4	4	4	4	4
	04		time constant of torque filter	50	100	50	50	50	100
	05	2nd	gain of position loop	500	500	500	500	500	500
	06		gain of velocity loop	1000	1000	1000	1000	1000	500
	07		time constant of velocity loop integration	500	5000	500	500	500	500
	08		filter of speed detection	4	4	4	4	4	4
	09		time constant of torque filter	50	50	50	50	50	50
	10	Velocity feed forward	gain	0	0	0	0	0	0
	11		filter	0	0	0	0	0	0
	12	Torque feed forward	gain	0	0	0	0	0	0
	13		filter	0	0	0	0	0	0
	14	2nd gain setup		0	1	0	0	0	0
	15	Position control switching	mode	0	0	0	0	0	0
	16		delay time	0	0	0	0	0	0
	17		level	0	0	0	0	0	0
	18		hysteresis	0	0	0	0	0	0
	19	Position gain switching time		0	0	0	0	0	0
	20	Velocity control switching	mode	0	5	0	0	0	0
	21		delay time	0	0	0	0	0	0
	22		level	0	5	0	0	0	0
	23		hysteresis	0	0	0	0	0	0
	24	Torque control switching	mode	0	0	0	0	0	0
	25		delay time	0	0	0	0	0	0
	26		level	0	0	0	0	0	1
	27		hysteresis	0	0	0	0	0	0

Parameter No.	Title		M1 (PullDown)	M2 (Jaw Rotation)	M3 (Pivot)	M4 (Film Roll)	M5 (BackSeal)	M7 (Shaker)	M9 (Poker)
Class	No.								
[Class 2] Damping control	00	Adaptive filter mode setup	1	0	1	1	1	1	4
	01	1st notch	frequency	1000	220	580	5000	5000	5000
	02		width selection	2	2	2	2	2	2
	03		depth selection	0	30	0	0	0	0
	04	2nd notch	frequency	1000	5000	1600	1000	1000	5000
	05		width selection	2	2	2	2	2	2
	06		depth selection	0	0	0	0	0	0
	07	3rd notch	frequency	5000	5000	5000	5000	5000	2456
	08		width selection	2	2	2	2	2	2
	09		depth selection	0	0	0	0	0	0
	10	4th notch	frequency	5000	5000	5000	5000	5000	5000
	11		width selection	2	2	2	2	2	2
	12		depth selection	0	0	0	0	0	0
	13	Selection of damping filter switching	0	0	0	0	0	0	0
	14	1st damping	frequency	0	0	0	0	0	0
	15		filter setup	0	0	0	0	0	0
	16	2nd damping	frequency	0	0	0	0	0	0
	17		filter setup	0	0	0	0	0	0
	18	3rd damping	frequency	0	0	0	0	0	0
	19		filter setup	0	0	0	0	0	0
	20	4th damping	frequency	0	0	0	0	0	0
	21		filter setup	0	0	0	0	0	0
	22	Positional command	smoothing filter	2	2	2	2	2	2
	23		FIR filter	0	0	0	0	0	0
[Class 3] Velocity/Torque/ Full-closed control	00	Speed setup, Internal/External switching	0	0	0	0	0	0	0
	01	Speed command	rotational direction selection	0	0	0	0	0	0
	02		input gain	500(*)	250	400	500(*)	500(*)	300
	03		reversal input	1	0	0	1	0	1
	04	Speed setup	1st	0	0	0	0	0	0
	05		2nd	0	0	0	0	0	0
	06		3rd	0	0	0	0	0	0
	07		4th	0	0	0	0	0	0
	08		5th	0	0	0	0	0	0
	09		6th	0	0	0	0	0	0
	10		7th	0	0	0	0	0	0
	11		8th	0	0	0	0	0	0
	12	Time setup	acceleration	0	0	0	0	0	0
	13		deceleration	0	0	0	0	0	0
	14	Sigmoid acceleration/deceleration time setup		0	0	0	0	0	0
	15	Speed zero-clamp	function selection	1	0	0	1	1	1
	16		level	30	30	30	30	30	30
	17	Torque command	selection	0	0	1	0	0	0
	18		direction selection	0	0	0	0	0	0
	19		input gain	30	30	30	30	30	30
	20		input reversal	0	0	1	0	0	0
	21	Speed limit value	1	0	0	0	0	0	0
	22		2	0	0	0	0	0	0
	23	External scale	selection	2	2	2	2	2	2
	24		numerator of division	0	0	0	0	0	0
	25		denominator of division	10000	10000	10000	10000	10000	10000
	26		reversal of direction	0	0	0	0	0	0
	27		Z phase disconnection detection disable	0	0	0	0	0	0
	28	Hybrid deviation	excess setup	122	1600	1600	122	122	1600
	29		clear setup	0	0	0	0	0	0

Parameter No.	Title		M1 (PullDown)	M2 (Jaw Rotation)	M3 (Pivot)	M4 (Film Roll)	M5 (BackSeal)	M7 (Shaker)	M9 (Poker)
Class	No.								
[Class 4] I/F monitor setting	00	Input selection	SI1 (Pin No.8)	8553090	8553090	8553090	8553090	8553090	8553090
	01		SI2 (Pin No.9)	8487297	8487297	8487297	8487297	8487297	8487297
	02		SI3 (Pin No.26)	9539850	9539850	9539850	9539850	9539850	9539850
	03		SI4 (Pin No.27)	394758	394758	394758	394758	394758	394758
	04		SI5 (Pin No.28)	4108	4108	4108	4108	4108	4108
	05		SI6 (Pin No.29)	197379	197379	197379	197379	197379	197379
	06		SI7 (Pin No.30)	3847	3847	3847	3847	3847	3847
	07		SI8 (Pin No.31)	263172	263172	263172	263172	263172	263172
	08		SI9 (Pin No.32)	328965	328965	328965	328965	328965	328965
	09		SI10 (Pin No.33)	3720	3720	3720	3720	3720	3720
	10	Output selection	SO1 (Pin No.10, 11 Line driver output)	197379	197379	197379	197379	197379	197379
	11		SO2 (Pin No.34, 35 Line driver output)	131586	131586	131586	131586	131586	131586
	12		SO3 (Pin No.36, 37 Line driver output)	65793	65793	65793	65793	65793	65793
	13		SO4 (Pin No.38, 39 Line driver output)	328964	328964	328964	328964	328964	328964
	14		SO5 (Pin No.12 Open collector output)	460551	460551	460551	460551	460551	460551
	15		SO6 (Pin No.40 Open collector output)	394758	394758	394758	394758	394758	394758
	16	Analog monitor 1	type	0	0	0	0	0	0
	17		output gain	500	500	500	500	500	500
	18	Analog monitor 2	type	4	4	4	4	4	4
	19		output gain	33	33	33	33	33	33
	20	Type of digital monitor		0	0	0	0	0	0
	21	Analog monitor output setup		0	0	0	0	0	0
	22	Analog input 1 (AI1)	offset setup	0	0	0	0	0	0
	23		filter	0	0	0	0	0	0
	24		overvoltage setup	0	0	0	0	0	0
	25	Analog input 2 (AI2)	offset setup	0	0	0	0	0	0
	26		filter	0	0	0	0	0	0
	27		overvoltage setup	0	0	0	0	0	0
	28	Analog input 3 (AI3)	offset setup	0	0	0	0	0	0
	29		filter	0	0	0	0	0	0
	30		overvoltage setup	0	0	0	0	0	0
	31	Positioning complete (In-position)	range	1	10	10	1	1	10
	32		output setup	0	0	0	0	0	0
	33	INP hold time		0	0	0	0	0	0
	34	Zero-speed		50	50	50	50	50	50
	35	Speed coincidence range		50	50	50	50	50	50
	36	At-speed (Speed arrival)		1000	1000	1000	1000	1000	1000
	37	Mechanical brake action	at stalling setup	0	0	0	0	0	0
	38		at running setup	0	0	0	0	0	0
	39	Mechanical brake action at running setup		30	30	30	30	30	30
	40	Selection of alarm output	1	0	0	0	0	0	0
	41		2	0	0	0	0	0	0
	42	2nd Positioning complete (In-position) range		10	10	10	10	10	10

Parameter No.	Title		M1 (PullDown)	M2 (Jaw Rotation)	M3 (Pivot)	M4 (Film Roll)	M5 (BackSeal)	M7 (Shaker)	M9 (Poker)
Class	No.								
[Class 5] Enhancing setting	00	2nd	0	0	0	0	0	0	0
	01	3rd	0	0	0	0	0	0	0
	02	4th	0	0	0	0	0	0	0
	03	Denominator of pulse output division	0	0	0	0	0	0	0
	04	Over-travel inhibit input setup	1	1	1	1	1	1	1
	05	Sequence at over-travel inhibit	0	0	0	0	0	0	0
	06	Sequence at Servo-Off	3	0	0	3	3	3	3
	07	Main power OFF	3	0	0	3	3	3	3
	08	sequence	3	0	0	3	3	3	3
	09	LV trip selection	1	1	1	1	1	1	1
		detection time	70	70	70	70	70	70	70
	10	Sequence at alarm	3	0	0	3	3	3	3
	11	Torque setup for emergency stop	0	0	0	0	0	0	0
	12	Over-load level setup	0	0	0	0	0	0	0
	13	Over-speed level setup	0	0	0	0	0	0	0
	14	Motor working range setup	10	10	10	10	10	10	10
	15	I/F reading filter	0	0	0	0	0	0	0
	16	Alarm clear input setup	0	0	0	0	0	0	0
	17	Counter clear input mode	3	3	3	3	3	3	3
	18	Command pulse	1	1	1	1	1	1	1
	19	inhibit input	4	4	4	4	4	4	4
		invalidation	1	1	1	1	1	1	1
		reading setup	4	4	4	4	4	4	4
	20	Position setup unit select	0	0	0	0	0	0	0
	21	Selection of torque limit	1	1	1	1	1	1	1
	22	2nd torque limit	300	300	250	300	300	300	300
	23	Torque limit	0	0	0	0	0	0	0
	24	switching setup	0	0	0	0	0	0	0
	25	External input	300	300	250	300	300	300	500
	26	positive direction torque limit	300	300	250	300	300	300	500
		negative direction torque limit	300	300	250	300	300	300	500
	27	Input gain of analog torque limit	30	30	30	30	30	30	30
	28	LED initial status	1	1	1	1	1	1	1
	29	RS232	2	2	2	2	2	2	2
	30	RS485	2	2	2	2	2	2	2
	31	Axis address	1	1	1	1	1	1	1
	32	Command pulse input maximum setup	4000	4000	4000	4000	4000	4000	4000
	33	Pulse regenerative output limit setup	0	0	0	0	0	0	0
	34	For manufacturer's use	4	4	4	4	4	4	4
	35	Front panel lock setup	0	0	0	0	0	0	0

Parameter No.	Title		M1 (PullDown)	M2 (Jaw Rotation)	M3 (Pivot)	M4 (Film Roll)	M5 (BackSeal)	M7 (Shaker)	M9 (Poker)
Class	No.								
[Class 6] Special setting	00	Analog torque feed forward conversion gain	0	0	0	0	0	0	0
	02	Velocity deviation excess setup	0	0	0	0	0	0	0
	04	JOG trial run command speed	300	300	300	300	300	300	50
	05	Position 3rd gain	0	0	0	0	0	0	0
	06	valid time	0	0	0	0	0	0	0
		scale factor	100	100	100	100	100	100	100
	07	Torque command additional value	0	0	0	0	0	0	0
	08	Positive direction	0	0	0	0	0	0	0
	09	Negative direction	0	0	0	0	0	0	0
	10	Function expansion setup	0	0	0	0	0	0	0
	11	Current response setup	100	100	100	100	100	100	100
	13	2nd Inertia ratio	250	250	250	250	250	250	250
	14	Emergency stop time at alarm	200	200	200	200	200	200	200
	15	2nd over-speed level setup	0	0	0	0	0	0	0
	17	Front panel parameter writing selection	0	0	0	0	0	0	0
	18	Power-up wait time	0	0	0	0	0	0	0
	19	Encoder Z phase setup	0	0	0	0	0	0	0
	20	Z-phase setup of external scale	0	0	0	0	0	0	0
	21	Serial absolute external scale Z phase setup	0	0	0	0	0	0	0
	22	A, B phase external scale pulse output method selection	0	0	0	0	0	0	0
	23	Disturbance torque compensating gain	0	0	0	0	0	0	0
	24	Disturbance observer filter	53	53	53	53	53	53	53
	27	Alarm latch time selection	1	5	5	1	1	1	1
	31	Real time auto	1	1	1	1	1	1	
	32	tuning	0	0	0	0	0	0	0
	34	Hybrid vibration	0	0	0	0	0	0	0
	35	suppression	10	10	10	10	10	10	10
	37	Oscillation detection level	0	0	0	0	0	0	0
	38	Alarm mask setup	0	0	4	0	0	0	0
	39	For manufacturer's use	0	0	0	0	0	0	0

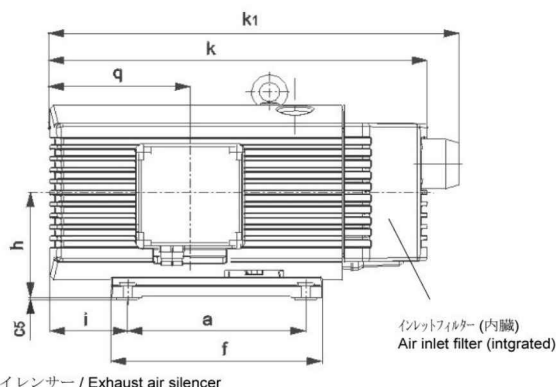
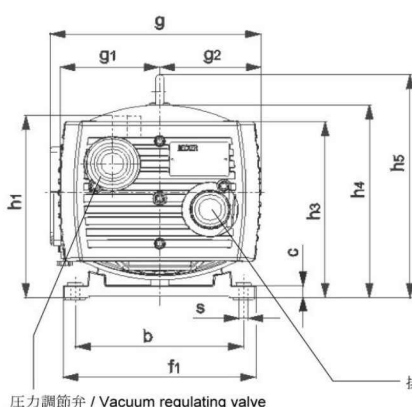
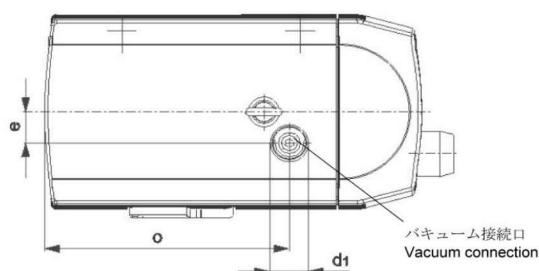
5.11 Vacuum Pump Parameter

5.11.1 Vacuum Pump Parameter (Becker)



ロータリーベーン式バキュームポンプ, オイルフリー, 空冷
Rotary vane vacuum pumps, oil-free, air-cooled

型式
VT 4.10
VT 4.16
VT 4.25
VT 4.40



型式 Type	吸引エア量 Suction air rate				到達真空度 Vacuum		モーター容量* Motor capacity installed*						電源電圧* Motor voltage*				回転数 Speed [RPM]		運転音 Noise level				重量 Weight
	Max 最大値						[kW]						3 相						[dB(A)]				
	[m³/h]												[V]										
	50 Hz		60 Hz		50 Hz		60 Hz		50 Hz		60 Hz		50 Hz		60 Hz		50 Hz		60 Hz				
VT 4.10	10		12		150		0.37				0.45		175-260/ 300-450	202-300/ 350-520		1420		1700	60		62		16,0
VT 4.16	16		19		150		0.55				0.70		175-260/ 300-450	202-300/ 350-520		1420		1700	61		64		22,4
VT 4.25	25		30		150		0.75				0.90		190-255/ 330-440	190-290/ 330-500		1420		1700	62		67		26,0
VT 4.40	40		48		150		1.25				1.50		190-255/ 330-440	190-290/ 330-500		1420		1700	67		72		38,5
Typ Type Tipo	a	b	c	c5	d1	e	f	f1	g	g1	g2	h	h1	h3	h4	h5	i	k	k1	o	q	ø s	
VT 4.10	160	112	15	3	G 1/2 "	35	200	142	206	90	90	107	169	176	189	195	106	387	429	257	123	7	
VT 4.16	202	125	15	3	G 1/2 "	35	242	155	231	102,5	102,5	113	188	186	205	211	73	416	452	291,5	151,5	7	
VT 4.25	220	190	15	3	G 3/4 "	40	260	238	260	125	125	140	226	227	250	290	96	465	505	302,5	173	7	
VT 4.40	220	208	15	3	G 3/4 "	40	260	238	280	125	125	140	226	227	250	290	131	532	572	363,5	192	7	

寸法単位 mm / Measures in mm * 単相仕様はお尋ね下さい / A.C. on request
仕様は予告無く変更することがあります。 / Right of modification reserved (17.04.03)

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Betriebsanleitung
 Operating Instructions
 Instructions de service
 Istruzioni d'uso
 Handleiding
 Instrucciones para el manejo
 Manual de instruções
 Naudojimosi instrukcija
 Kasutusjuhend
 Lietošanas instrukcija
 Οδηγίες χρήσης
 取扱説明書
 사용설명서

Driftsinstruks
 Driftsinstruktioner
 Käyttöohje
 Driftsvejledning
 Instrukcja obsługi
 Kezelési útmutató
 Návod k obsluze
 Navodilo za uporabo
 Návod na obsluhu
 El Kitabı
 Инструкция по эксплуатации
 使用说明书

VT 4.40

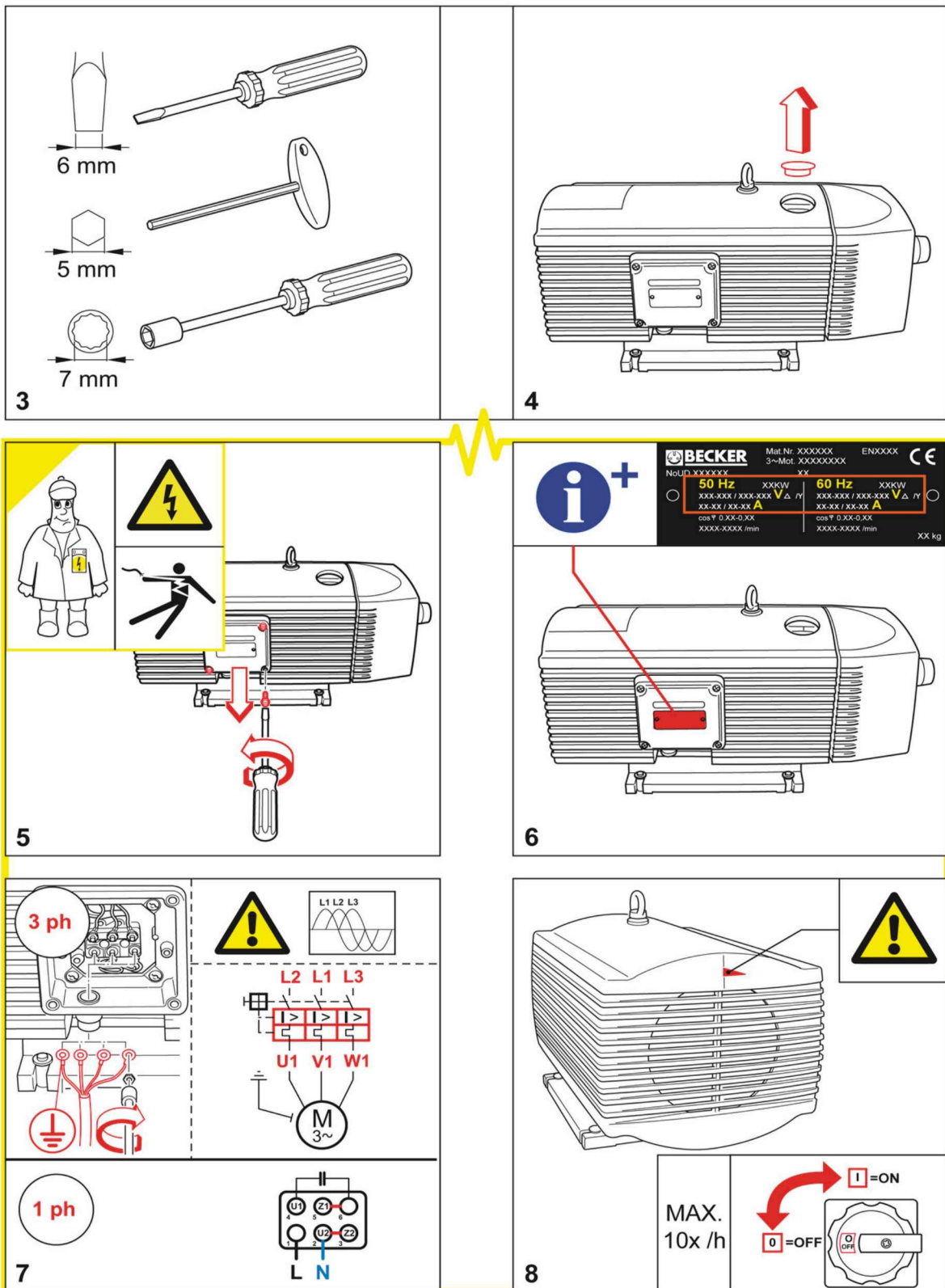
98/37 EG
73/23 EWG



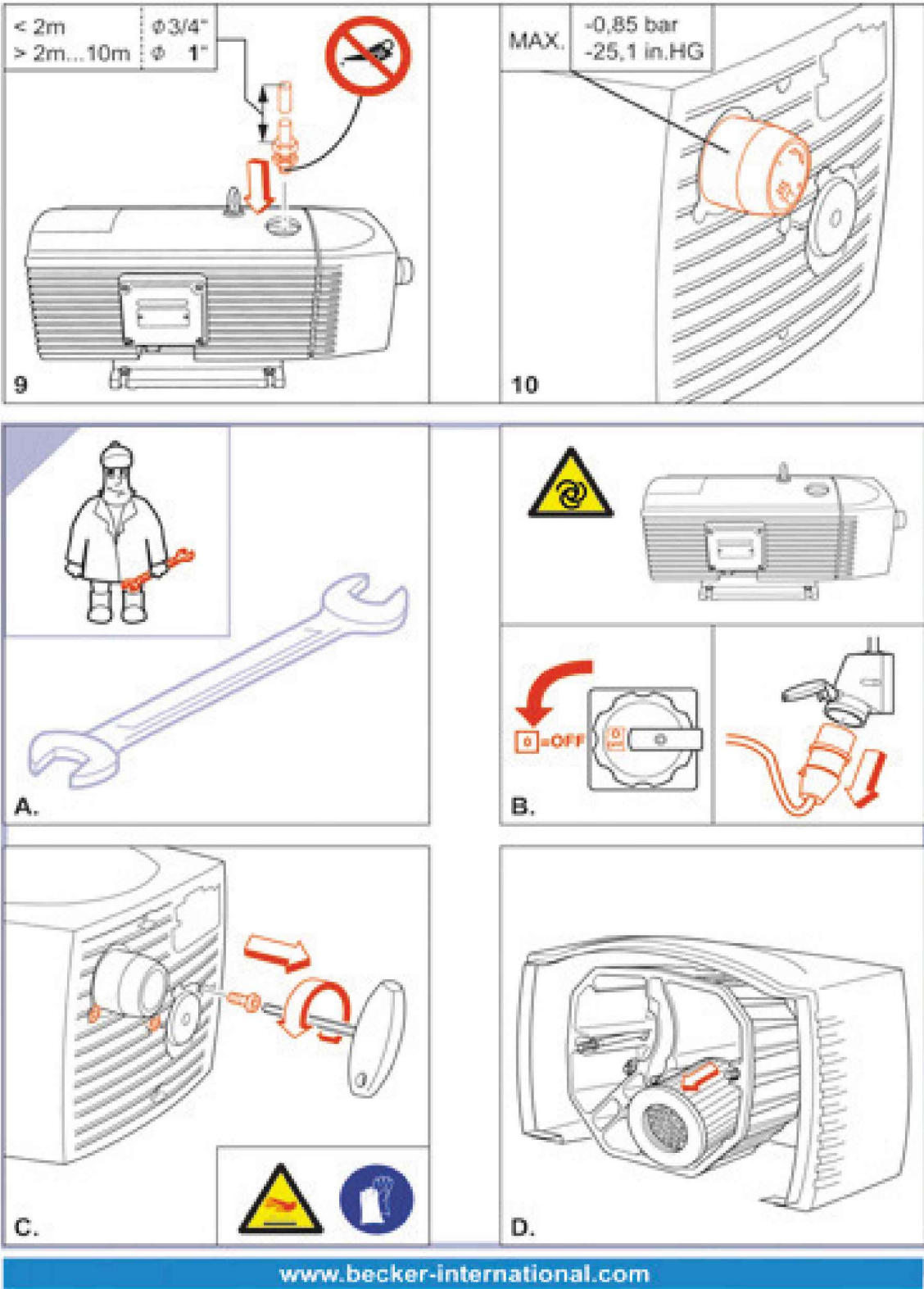
 			 MAX. VACUUM		mbar
			 MAX.		m³/h
				$L_{pA} = 67 \text{ dB(A)} - 50\text{Hz}$ $L_{pA} = 72 \text{ dB(A)} - 60\text{Hz}$ $K_{pA} = 3 \text{ dB(A)}$	

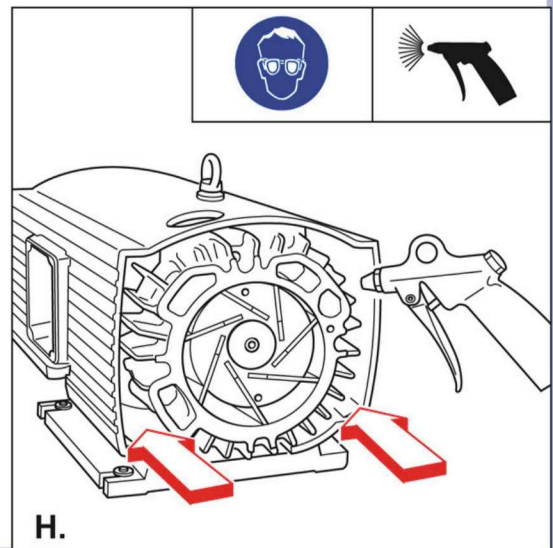
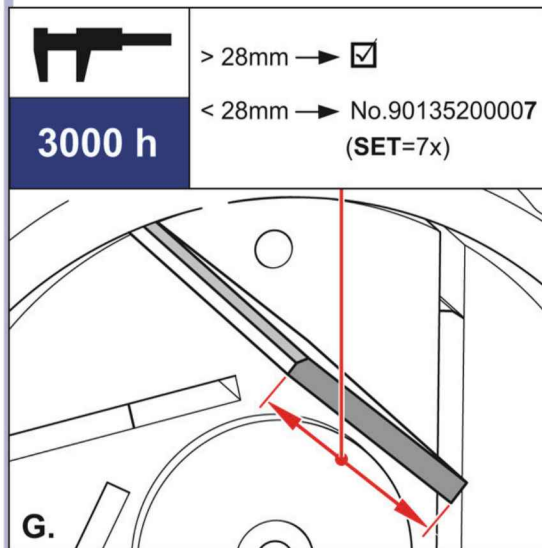
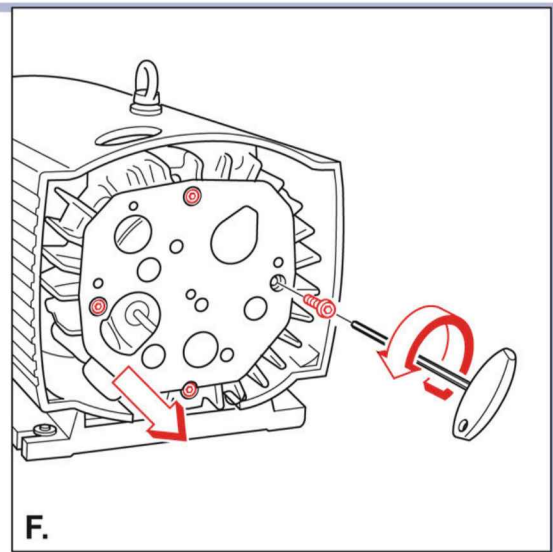
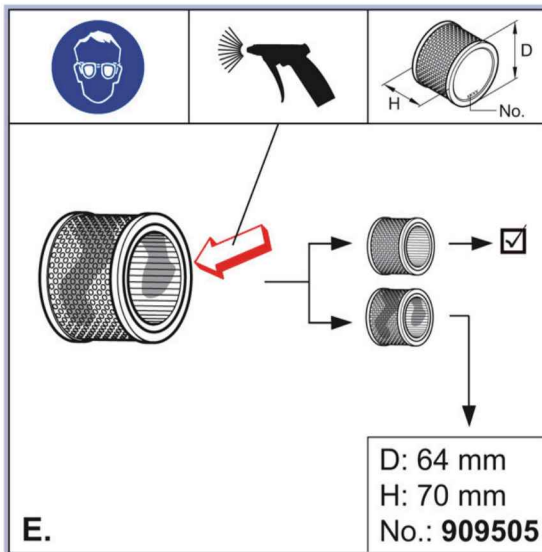
	$A > 100\text{mm}$ $A > 4"$	$> 5^\circ\text{C}/41^\circ\text{F}$ $< 45^\circ\text{C}/113^\circ\text{F}$	$\text{max. } 90\%$
1 39 kg 85 lbs			
	2		

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Gebr. Becker GmbH
Hölker Feld 29-31
D-42279 Wuppertal

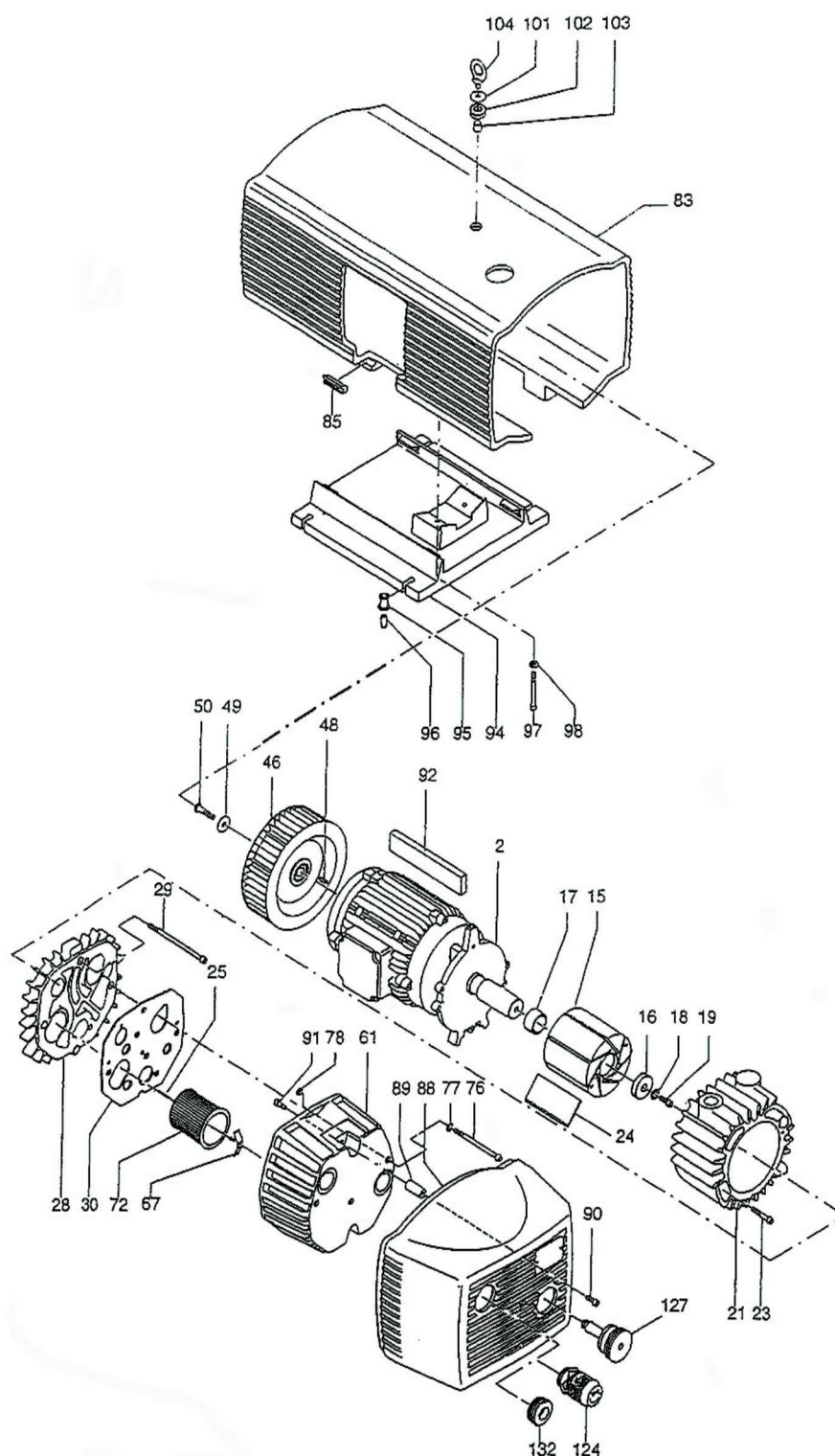
info@becker-international.com

(D) Service:

Tel: +49 (0)202 697-171

Fax: +49 (0)202 64 44 74

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Pos Beschreibung

15	ROTOR
16	CLAMPING DISC
17	STAR-TOLERANCE-RING
18	TOOTHED SPRING WASHER
19	SOCKET HEAD SCREW
21	PUMP BODY
23	SOCKET HEAD SCREW
24	CARBON VANES
25	LOCATING PEG
28	LID
29	SOCKET HEAD SCREW
30	GASKET
46	FAN
48	KEY
49	WASHER
50	DRIVE SHAFT
61	FILTER COVER
67	LEAF SPRING
72	FILTER CARTRIDGE
76	SOCKET HEAD SCREW
77	SEALING RING
78	O-RING
83	PROTECTING HOOD
84	ANTIVIBRATING STRIP
85	RUBBER BUFFER
88	PROTECTING HOOD
89	RUBBER BUFFER
90	SOCKET HEAD SCREW
92	EDGE PROTECTION
94	BASE
95	RUBBER BUFFER
96	SPACER BLOCK
97	SOCKET HEAD SCREW
98	WASHER
101	SOCKET HEAD SCREW
102	SPACER BLOCK
103	SPACER TUBE
104	RING SCREW
124	VACUUM REGULATING VALVE
127	BLOW-OFF VALVE
132	LOCKING SCREW

5.12 Digital Pressure Switch

This section describes the default value of the vacuum pump parameter.

NOTE

- Set the vacuum pressure in operation to 40 kPa.

Setting item		Setting value
Default setting	Setting Display Color	SoG (ON : green)
	Setting Operation Mode	HYS (Hysteresis mode)
	Setting Output Form	nC (normal close)
	Setting Response Time	640 (640ms)
	Setting Auto Preset	mAn (manual)
Setting Pressure Setting (n-1)		-10kPa

Setting pressure unit: kPa (Europe)
psi (North America)

Names and Functions of Individual Parts

Main Unit

Indication Light : Displays switch operation condition.

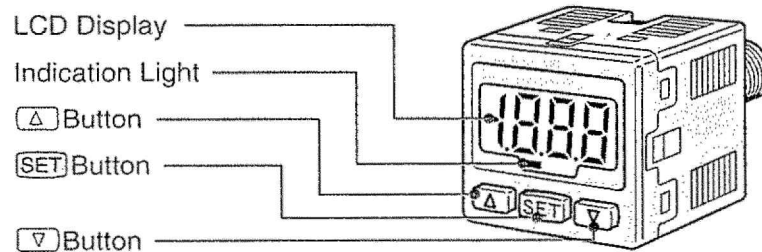
(Green Light)

LCD Display : Displays the current status of pressure, setting mode, selected indication unit and error code. Four display modes can be selected: display always in red or green only, or changing from green to red linked to output.

 Button : Alters the mode or increases ON/OFF set value.
Press this button to change to the peak display mode.

 Button : Alters the mode or decreases ON/OFF set value.
Press this button to change to the bottom display mode.

 Button : Press this button to change to either mode and to set a set value.



Setting

Setting Procedures

Measurement Mode

Initialize

Set output mode, response time and display color change.

Pressure Setting

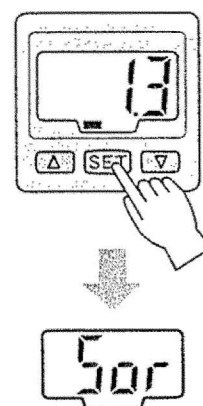
Input a set value for pressure to perform switch output.

Measurement Mode

Detects pressure, displays values and performs switching.
Other functions such as zero clear can also be set if necessary.

Initialize

Press the **[SET]** button continuously for more than two seconds. The display shown at the lower right will pop up to allow setting of a display color. In the case that the unit specification of model indication is M, the SI units will be fixed. If no symbol is supplied, see "Selecting Indication Unit".



1. Display Color Setting

Select a color for the LCD display.

When changing the display color, press the Δ or ∇ button to select a display color.



Sor (Red/ON) $\Leftrightarrow$ SoG (Green/ON) $\Leftrightarrow$ rEd (Red) $\Leftrightarrow$ Grn (Green)

Press the **SET** button to set the desired display color and to move on to setting a desired operation mode.

If the mode is set to analog output, press the Δ or ∇ button, to select a desired display color from

"Grn" (Green) $\Leftrightarrow$ "rEd" (Red), then press the **SET** button. Setting of a desired operation mode will become available.

2. Operation Mode Setting

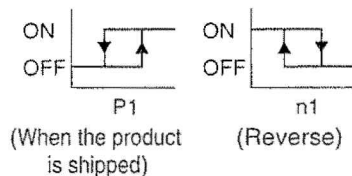
A desired switch operation mode can be selected.

The operation mode currently selected will be displayed. Select a desired operation mode by pressing the Δ or ∇ button.

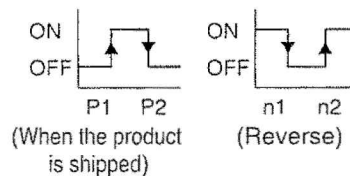


HyS $\Leftrightarrow$ wnd
(Hysteresis) (Window Comparator)

Hysteresis Mode



Window Comparator Mode



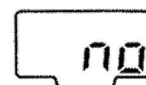
Press the **SET** button to move on to setting an output mode.

Setting (continue)**3. Output Mode Setting**

A desired output mode can be set freely for switch output.

The output mode currently selected will be displayed.

Press the Δ or ∇ button to switch to normal output "no" or reverse output "nC".



no $\Leftrightarrow$ nC
(Normally open) (Normally closed)

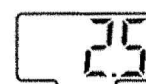
Press the **SET** button to move on to setting a desired response time.

4. Response Time Setting

A response time for switch output can be set as user desires.

Setting of a response time prevents chattering output.

The response time currently set will be displayed. Select a desired response time by pressing the Δ or ∇ button.



2.5 $\Leftrightarrow$ 20 $\Leftrightarrow$ 160 $\Leftrightarrow$ 640 $\Leftrightarrow$ 1280

If the operating mode is set to Hysteresis, press **SET** button to set and move on to setting Auto Preset mode.

If the operating mode is set to Window comparator, press **SET** button to set and return to Measurement mode.

5.Auto Preset Setting



This function is for memory of a measurement pressure as a reference value when Auto Preset input is set. The settings currently set will be displayed. Press the or button to set to Auto Preset.

mAn $\Leftrightarrow$ AUt
(Manual Setting) (Auto Preset)

Press the button to return to the set Measurement mode.

Selecting Indication Unit



If the unit specification of the model indication is without "M"

The indication unit can be selected freely.

Pressing the or button will change the unit and will automatically convert set values.

The units will change in the following order :

PA $\Leftrightarrow$ GF $\Leftrightarrow$ bAr $\Leftrightarrow$ PSi $\Leftrightarrow$ inH $\Leftrightarrow$ mmH

For Vacuum and Low Pressure

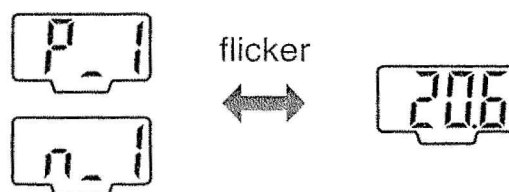
Pa $\Leftrightarrow$ kgf/cm² $\Leftrightarrow$ bar $\Leftrightarrow$ psi $\Leftrightarrow$ inchHg $\Leftrightarrow$ mmHg

For Positive Pressure

MPa $\Leftrightarrow$ kgf/cm² $\Leftrightarrow$ bar $\Leftrightarrow$ psi

Press the button to set and to move on to setting a display color.

If set to Manual Setting



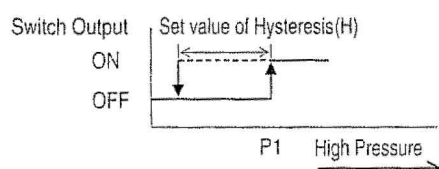
Press the SET button in the Measurement mode to display set values. "P_1" or "n_1" and the current set value will flicker alternately. Press the **SET** button to display the next set value. Press the **Δ** or **▽** button to enter into the Value Change mode. (See "Value Setting")

If Hysteresis mode is set

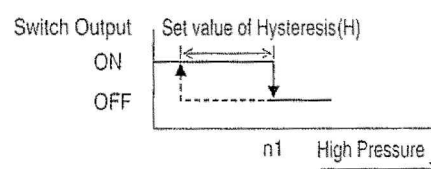


If the Hysteresis mode is set, "H" and the set value of Hysteresis will be displayed alternately after the setting for "P1" or "n1". Press the **SET** button to return to the normal Measurement mode. Press the **Δ** or **▽** button to enter into the Value Change mode. (See "Value Setting")

If set to Normally Open Mode

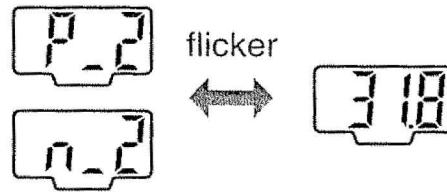


If set to Normally Closed Mode

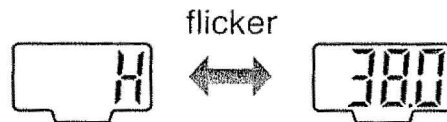


In case Hysteresis is set at less than or equal to 2 digits, switch output may chatter if input pressure fluctuates near the set point.

If window comparator mode is set



If the Window comparator mode is set, "P2" or "n2" and the current set value will be displayed alternately after the setting for "P1" or "n1". Press the **SET** button to display the next set value.
(Hysteresis : H)

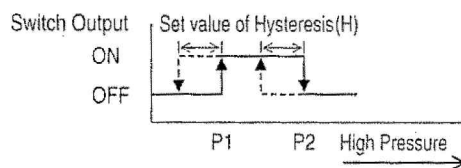


Press the **Δ** or **▽** button to enter into the Value Change mode.
(See "Value Setting")

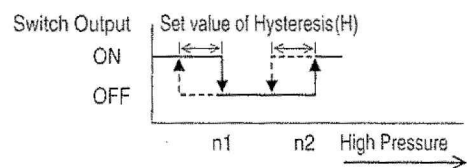
Next, "H" and the set value of Hysteresis will be displayed alternately. Press the **SET** button to return to the normal Measurement mode. Press the **Δ** or **▽** button to enter into the Value Change mode. (See "Value Setting")

If the initialize value is the Normally Open mode, "P_1" will be displayed. "n_1" will be displayed if it is Normally Closed mode.
The set pressure value can be checked without holding or stopping switch output operation.

If set to Normally Open Mode



If set to Normally Closed Mode



Pressure Setting (continue)**Fine Adjustment Mode
(Fine Adjustment Function of Display Value)**

Press the **SET** button and **▽** buttons simultaneously for longer than two seconds in the Measurement mode. "FSt" and current pressure Measurement value will be displayed. Press the **△** or **▽** button to change the set value. If no operation is made for longer than two seconds or press the **SET** button, the pressure switch will display the current pressure Measurement value which will then flicker alternately with "FSt".



Press the **SET** button to display an adjusted amount (percentage), which will then flicker alternately with "FSC". Press the **SET** button to return to the normal Measurement mode.

If set to Auto Preset Mode



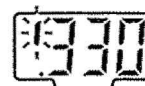
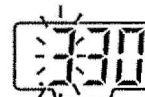
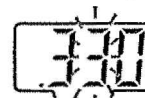
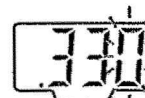
Press the **SET** button during the Measurement mode to ready the Auto Preset mode. "AP1" will be displayed. Setting pressure in this condition will ready the equipment. Press the **Δ** and **▽** buttons simultaneously while "AP1" is displayed to return to the Measurement mode.

To execute Auto Preset, press the **SET** button and "A1L" will be displayed. Perform adsorption and desorption operations. Detection will be made and a set value will be stored in the memory automatically. Press the **SET** button while "A1L" is displayed to finish setting and to return to the normal Measurement mode.

Value Setting

To input a value for pressure setting or other purposes:

1. Press the **Δ** or **▽** button to enter the Set Value Change mode. The first row will flicker.
2. Press the **Δ** or **▽** button to set a desired value.
(No operation within ten seconds after the Set Value Change mode was selected results in automatic setting of the value appearing in the display window and in changing of the mode from Set Value Change mode to Set Value Indication mode.)
3. Press the **SET** button to make the value one digit higher flicker.
(If the highest place is zero, "I" or "J" will flicker. "I" means "+zero", "J" means "-zero".)
(In the case that the **SET** button is pressed in the highest place, the first digit will flicker.)
4. Press the **SET** button continuously for longer than one second to return to displaying set values.

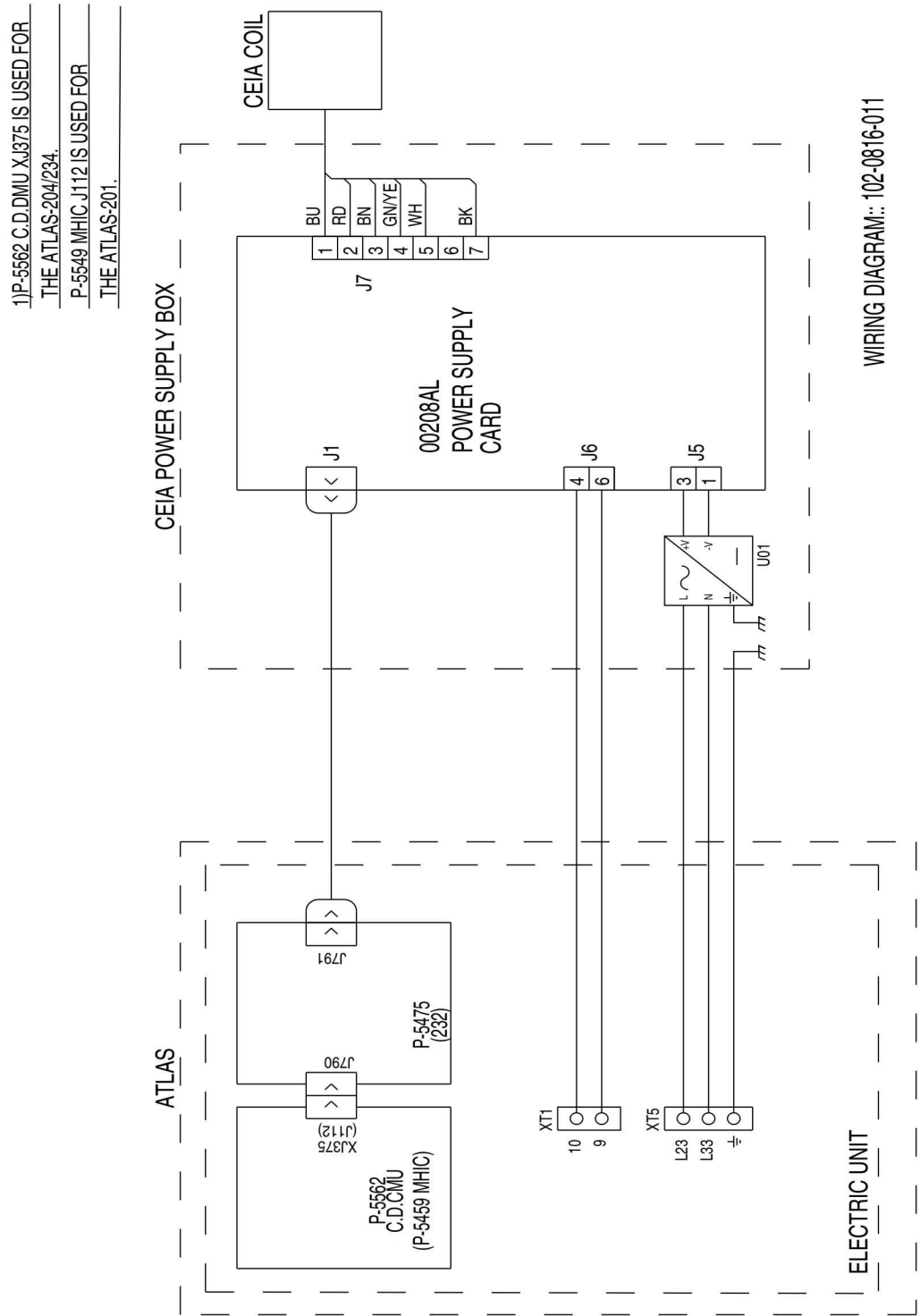


5.13 ITPS (OPTION)

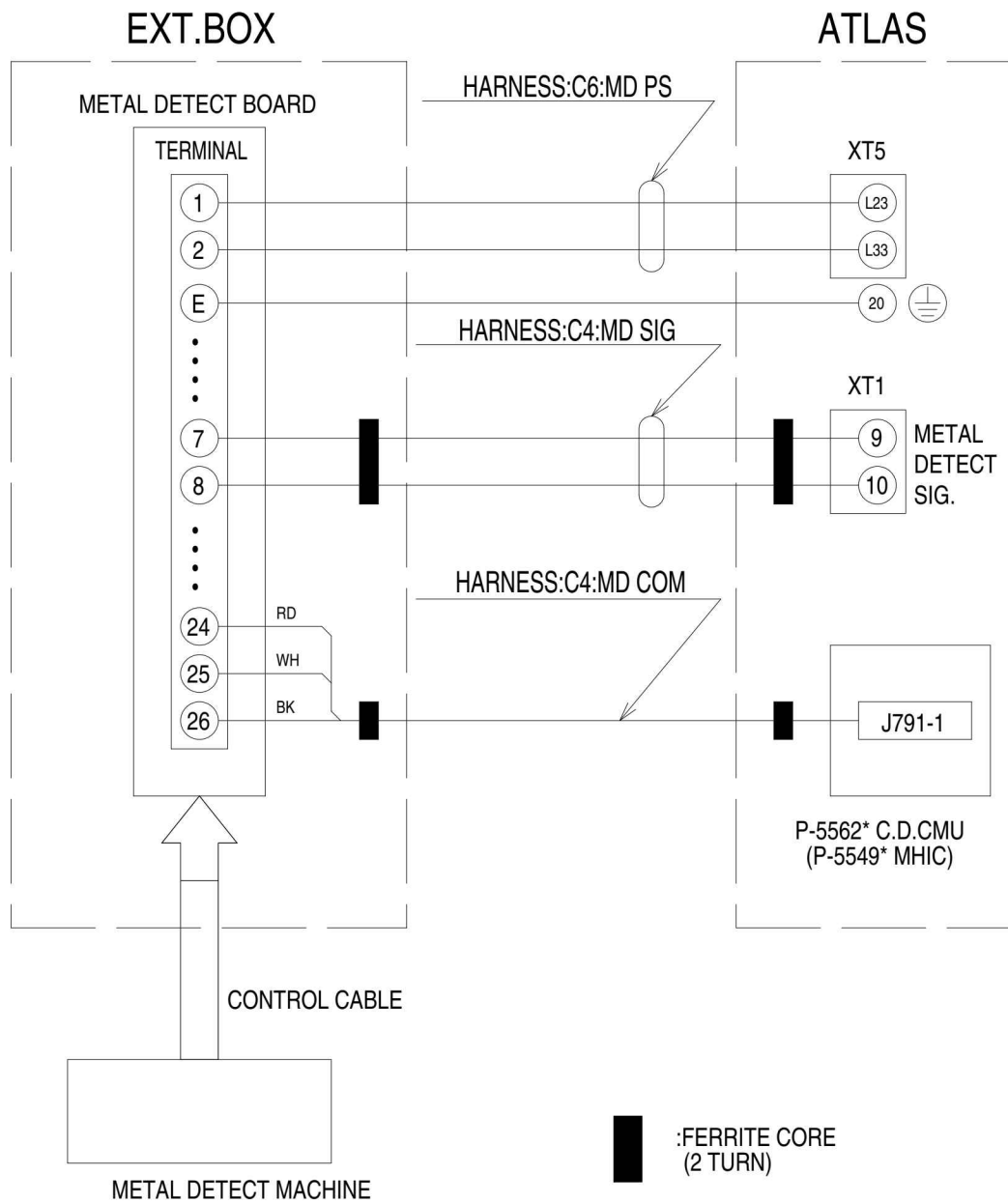
5.13.1 Block diagram



5.13.2 Block diagram

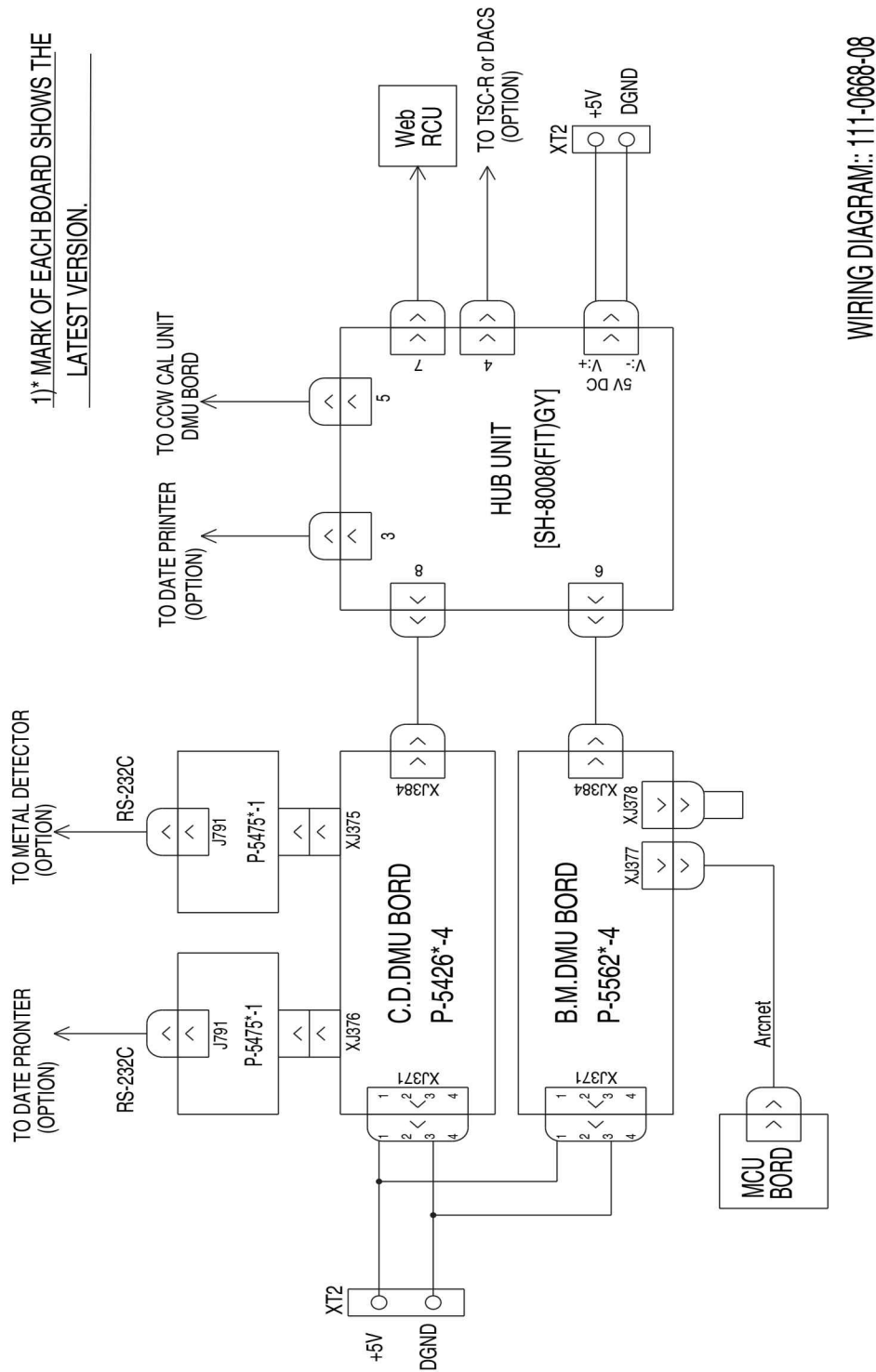


5.13.3 Metal Detect / ATLAS 204/234 integrated operation

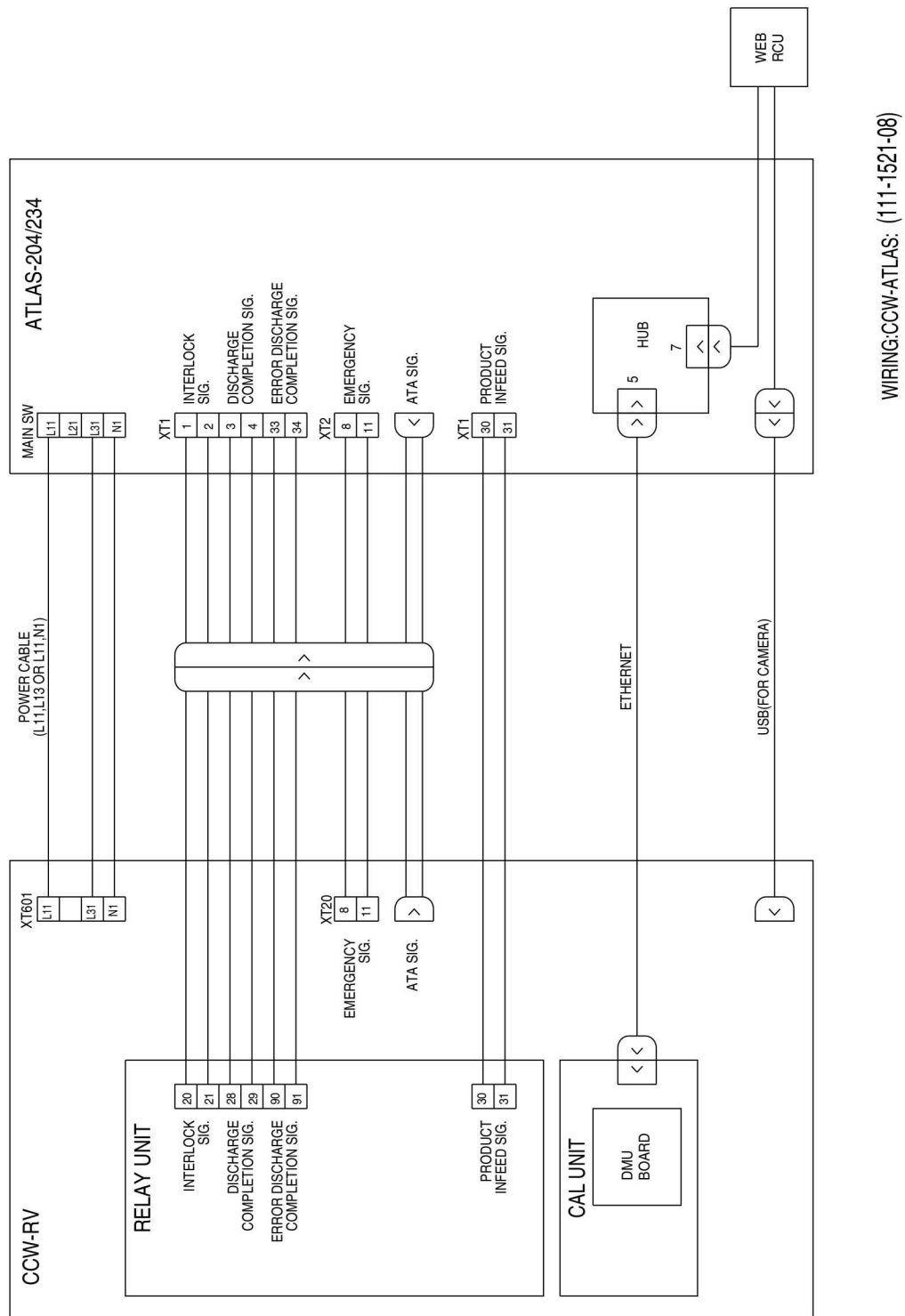


METAL DETECT/ATLAS INTEGRATED OPERATION:: 110-8666-02

5.13.4 Block diagram (option) [ITPS]



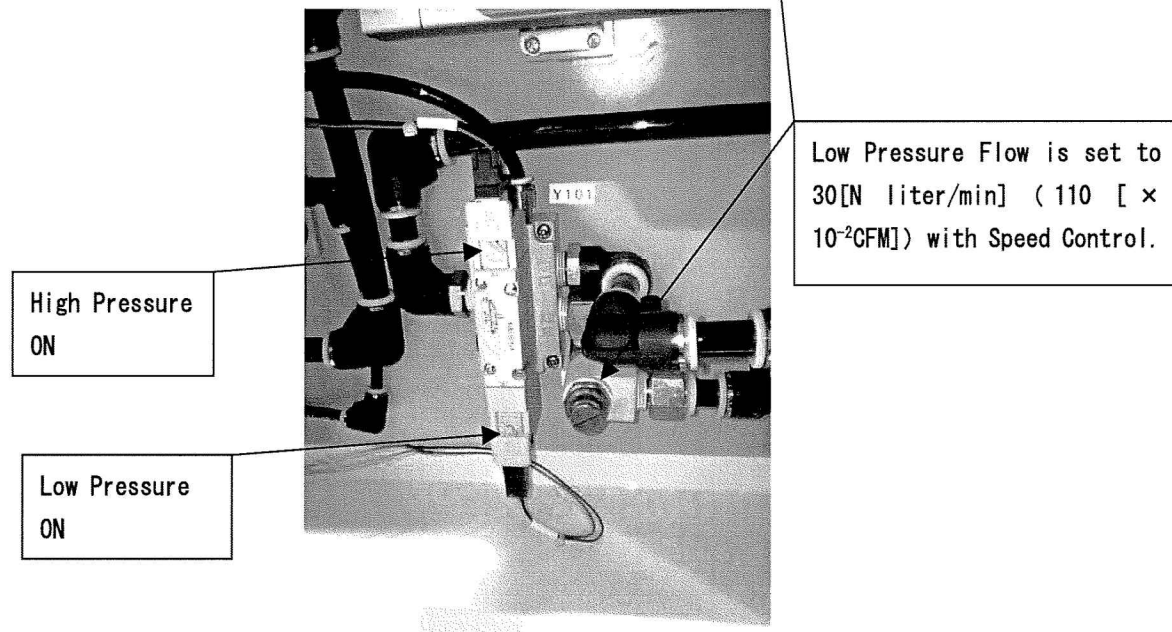
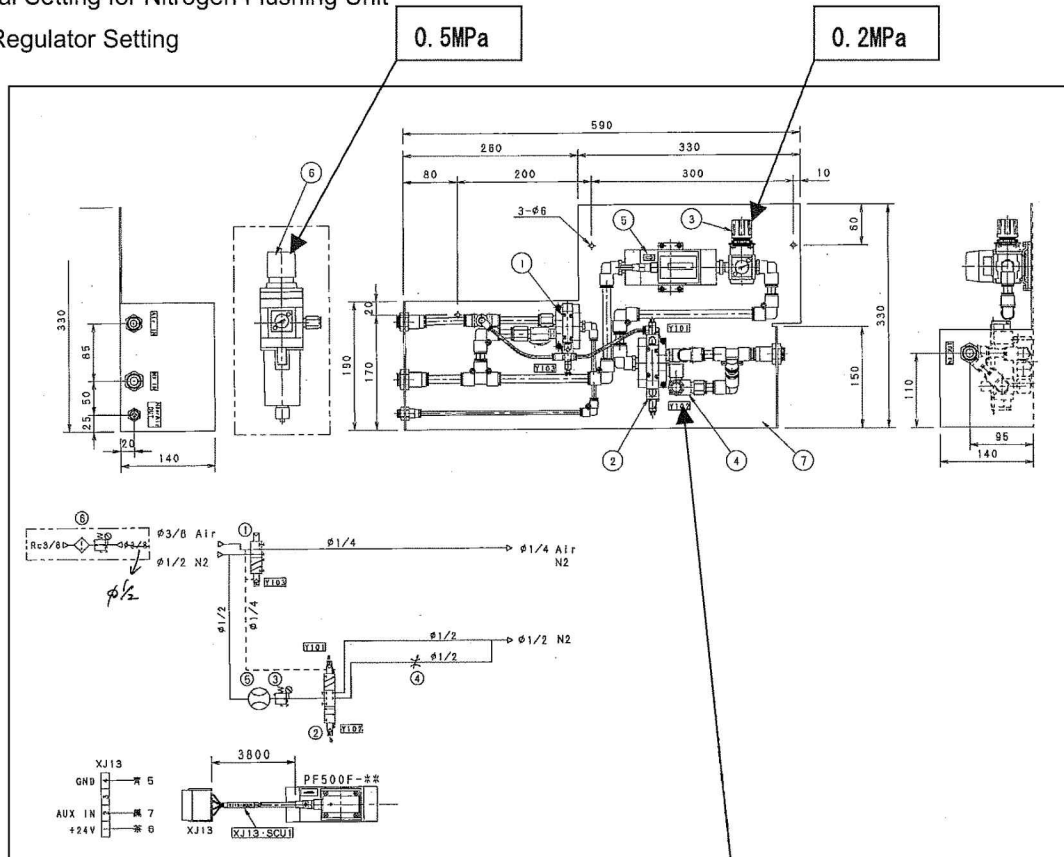
5.13.5 Block diagram : CCW-RV - ATLAS 204/234



5.14 ATLAS 204/234 Nitrogen Flushing Setting (Option)

Initial Setting for Nitrogen Flushing Unit

1) Regulator Setting



1)-2. Flow Meter Setting

How to set the alarm

After power is turned on, monitor is in Flow rate measurement status (Measurement mode) as fig. right. Description is given below to change the below settings in measurement mode.

P(Parameter):0, H(High):50, L(Low):20, h(Hysteresis):0

(All LEDs light up for 3 seconds for a hardware check, followed by flow range for 2 seconds, right after the power is turned on.)

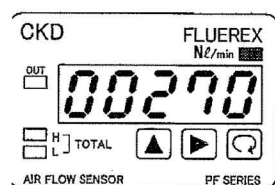


图 5-5

- 1) Changing from the measurement mode to the write mode

*Press the shift key  for 2 seconds.

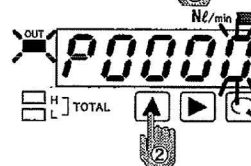
OUT LED blinks and "P" is shown at first place of the 7segment LED from the left, and the fifth place blinks.



- 2) Setting the parameter

*Press the up key .

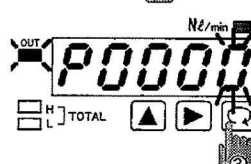
If parameter is "0", press the change key .



- 3) Shifting to the low setting

*Press the change key .

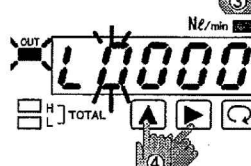
"L" is shown at the first place of 7segment LED from the left and the second place blinks.



- 4) Low setting

*Press the shift key  and the up key  to set the low value.

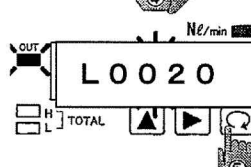
In case of LOW=20, press the shift key twice, and the up key twice.



- 5) Shifting to the High setting

*Press the change key .

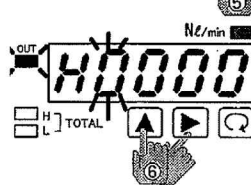
"H" is shown at the first place of 7segment LED from the left and the second place blinks.



- 6) High setting

*Press the shift key  and the up key  to set the high value.

In case of HIGH=50, press the shift key twice, and the up key 5 times.



× 10⁻²CFM

L 0 0 7 1

7) Shifting to the hysteresis setting

*Press the change key 

"H" is shown at the first place of 7segment LED from the left and the second place blinks.

8) Hysteresis Setting

*Press the shift key  and the up key  to set the hysteresis value. h(hysteresis) keeps "0".

9) Shifting to the cumulative value clear mode

*Press the change key 

Cumulative Flow is shown at 7segment LED and all place blink. Then, Unit display lamp is "L-TOTAL".

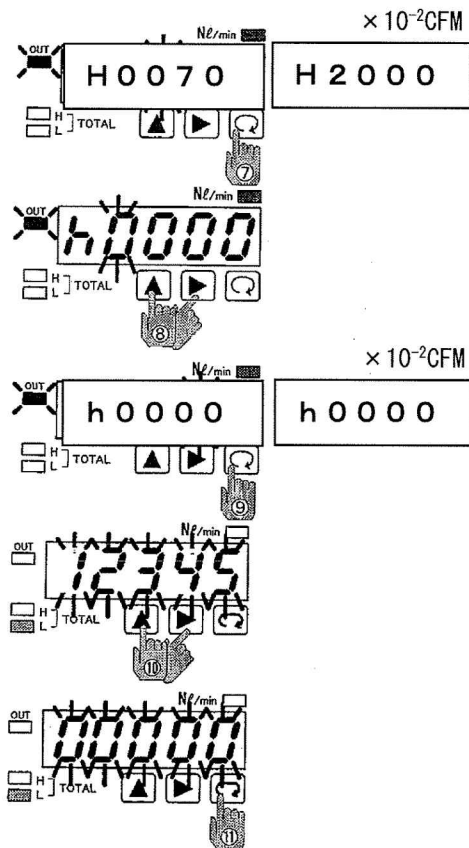
10) Clearing the cumulative value

*Press the shift key  and the up key  simultaneously for 5 seconds to clear the cumulative value.

[Notes]

If you do not clear the cumulative value, go to 11)Completion process.

- 11) *Press the change key  to complete the setting and return the measurement mode (the menu right after power on).



2) Flow Meter

Initial setting for Gas Flow Meter is as follows. Please set properly at your site.

(Measure to be used depends on used area.)

	Other area	North America
Measure	N liter/min	$\times 10^{-2} \text{CFM}$
P(Parameter)	0	0
L(Low)	2 0	7 1
H(High)	7 0	2 0 0 0
h(Hysteresis)	0	0
(Low Pressure Flow)	(3 0)	(1 1 0)

Conversion Formula: $1 \text{CFM} = 28.317 \text{N liter/min}$

Amendment list

Version	Date	Revision number	Description
1	March 2015	0001763592011	- Added M7 shaker parameter



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